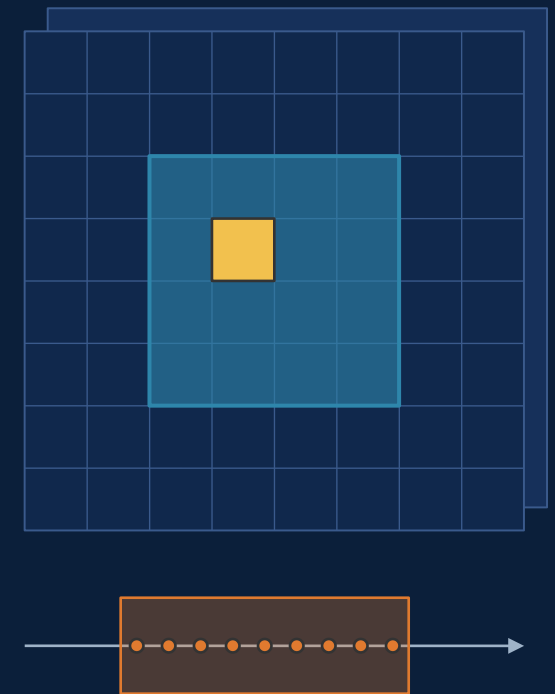


Geospatial representation models

A stencil-level overview of AlphaEarth, TESSERA, OlmoEarth, TerraMind, Prithvi-EO 2.0 and IBM's Prithvi-based ocean model (Granite-Geospatial-Ocean)

What area in space and what span of time goes into each embedding — and what each model actually hands you



Geospatial representation models

This deck compares six "representation models" for Earth observation, and then uses them to sharpen the design of our own model for the ocean.

A representation model (also called an embedding model or foundation model) is a neural network that is trained on huge amounts of raw data without labels, and whose job is to turn each place on Earth into a short list of numbers — a vector — that summarises what is going on there. That vector is called an embedding. Once you have good embeddings, many downstream tasks (mapping crops, detecting floods, estimating an ocean current) become much easier because a small model can be trained on top of the embeddings with very few labels.

The word "stencil" runs through the whole deck. It is our term for the piece of the world that feeds one embedding: how large an area around a pixel the model looks at, and how long a stretch of time it summarises. The little drawing on the right shows the idea — a grid of pixels with one output pixel (yellow) and its spatial context (blue), and a timeline with the observations that go into it (orange).

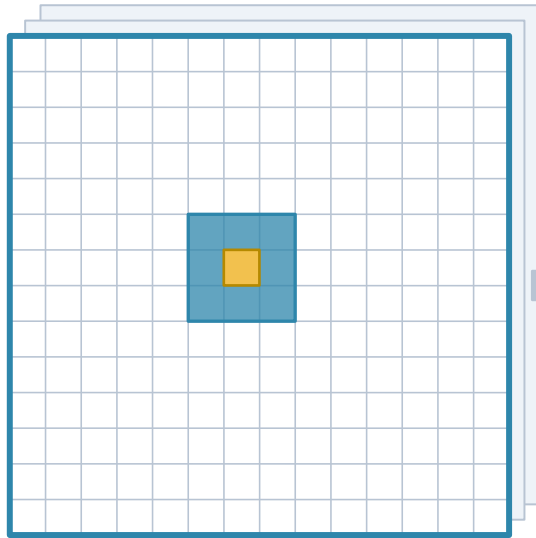
The six models are: AlphaEarth (Google DeepMind), TESSERA (University of Cambridge), OlmoEarth (Allen Institute for AI), TerraMind (IBM and ESA), Prithvi-EO 2.0 (IBM and NASA), and IBM's ocean model, Granite-Geospatial-Ocean. The last third of the deck applies what we learn from them to "Earth 2", our own ocean project.

How to read the stencil diagrams

Every model slide uses the same three-panel convention

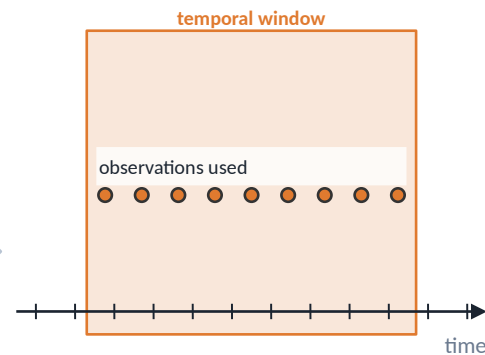
Stencil — what feeds one embedding

SPACE



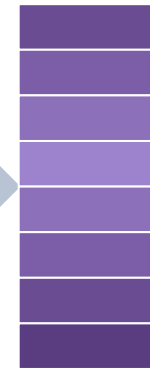
Grid = input tile the encoder sees at once (blue outline).
Blue square = one token / receptive field. Yellow = the output pixel.
Stacked sheets = co-registered modalities / channels.

TIME



Shaded band = window summarised by one embedding.
Dots = observations sampled inside it. A single tall bar = one snapshot acquisition.

OUTPUT



D dims

one vector per pixel, per token, or per tile

Four questions we ask of each model

- Per-pixel or patch? Does the embedding of a pixel see its neighbours (patch/ViT) or only its own time series (pixel-wise)?
- Snapshot or period summary? One acquisition, a handful of dates, or a whole year folded into one vector?
- Product or model? Do we get precomputed global embeddings, open weights we can run on our own grid, or both?
- Does it touch the ocean? Land-only masking is the norm; only one of the six is trained on sea pixels.

How to read the stencil diagrams

Every model slide uses the same three-panel picture, so this slide is the legend.

SPACE panel (left). The grid is the tile — the square block of pixels the model processes at once. The blue outline marks the tile edge. The small blue square is one "token": a token is the unit the neural network actually reasons about, usually a small patch of pixels (for example 16 by 16) squashed into one vector. The yellow square is the single output pixel we are asking about. The stacked sheets behind the grid stand for the different data sources — for example optical imagery, radar and elevation — that are laid on top of each other ("co-registered") so that every pixel has a value from each source.

TIME panel (middle). The horizontal axis is time. The shaded band is the "temporal window": the span of time that one embedding summarises. The dots are the individual satellite observations inside that window. If instead there is a single tall bar, the model looks at one moment only — a snapshot.

OUTPUT panel (right). The purple stack represents the embedding vector. "D" is its dimension, the number of entries in the vector — 64, 128, 768 and so on. The caption says whether you get one vector per pixel, per token, or per whole tile.

The four questions on the right are the ones we ask of every model. "Per-pixel or patch" asks whether the model uses neighbouring pixels at all. "Snapshot or period summary" asks whether time is folded into the vector. "Product or model" asks whether the makers hand you pre-computed embeddings, the trained network ("weights"), or both. And "does it touch the ocean" is our own concern: almost all of these models deliberately throw away sea pixels.

AlphaEarth Foundations (Google DeepMind)

An 'embedding field': every 10 m land pixel $\rightarrow$ 64-D unit vector per year, 2017–2024

Who / when DeepMind + Google Earth Engine; arXiv 2507.22291 (Jul 2025, v2 Sep 2025).

Inputs Sentinel-2, Landsat 8/9, Sentinel-1, PALSAR-2, GEDI lidar, ERA5-Land, GRACE, GLO-30 DEM, NLCD, plus text (Wikipedia, GBIF) — >3 B observations.

Architecture Space-Time-Precision (STP) encoder — 15 blocks, each with three operators at three resolutions (spatial attention @ L/16, time-axial attention @ L/8, 3×3 convs @ L/2; widths 1024 / 512 / 128) exchanging state through a learned Laplacian pyramid (next slide). Teacher/student video embedders + text alignment; ~480 M deployed (1 B trained). Losses: reconstruction, batch-uniformity on S^{63} , consistency, text-contrastive.

Stencil Output 10 m pixel. Input frame 1.28 km × 1.28 km (128 × 128 px; supplement S2.1/S8.2): inference runs on 960 m tiles buffered by 160 m, so a pixel's context is at most the 1.28 km frame, with spatial attention global inside it. Continuous 'valid period' $[t_0, t_1]$ — arbitrary windows in principle; released product = calendar-year summaries.

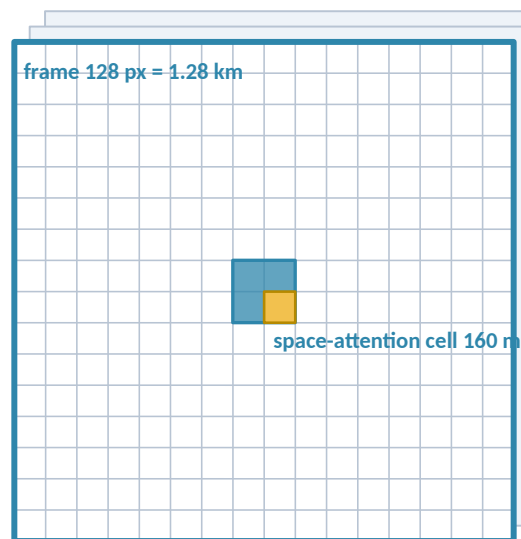
Output 64-D, unit-norm, int8 in the public dataset; ~1.4 T footprints / yr. Land + coastal/inland water; no open ocean.

Availability Dataset only (CC-BY 4.0) in Earth Engine / GCS. Weights and code not released.

Results ~23.9 % average error reduction vs. prior featurizations over 15 evaluations (10.4 % at 10-shot, 4.2 % at 1-shot); evapotranspiration regression R^2 0.58. Per-dataset scores are charts only.

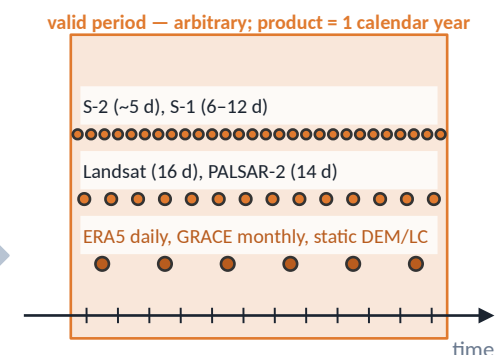
Stencil — what feeds one embedding

SPACE



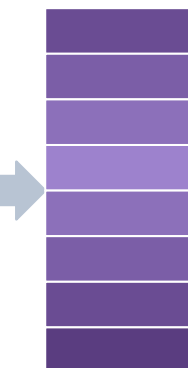
Output pixel: 10 m × 10 m (drawn 8× too large). Context: the 1.28 km frame. Spatial attention is global inside it on an 8 × 8 grid of 160 m cells; the outer 80 m is trimmed at inference, so no pixel sees further than ~640 m. ~10 co-registered sources at 10 m ... 300 km native resolution.

TIME



All observations in the window are summarised — a period embedding, not a snapshot. Model supports interpolation/extrapolation in time; only annual layers are public.

OUTPUT



64-D
per 10 m pixel per year
int8, unit sphere

AlphaEarth Foundations (Google DeepMind)

AlphaEarth Foundations is Google DeepMind's model, published in July 2025. Its central idea is the "embedding field": every 10-metre pixel of land on Earth gets a 64-number vector for every year from 2017 to 2024, and Google publishes those vectors as a dataset inside Google Earth Engine.

Inputs: a very wide mix — optical imagery (Sentinel-2, Landsat), radar (Sentinel-1, PALSAR-2), laser altimetry (GEDI), climate reanalysis (ERA5-Land), gravity (GRACE), elevation, land cover and even text descriptions. Over three billion individual observations went in.

Architecture: the encoder is called Space-Time-Precision, STP, and has 15 blocks. The next four slides walk through it. The training uses several losses at once: reconstructing masked-out inputs, keeping the embeddings spread evenly over a sphere ("batch uniformity"), consistency between a teacher and a student network, and matching to text. The deployed network has about 480 million parameters.

The stencil is the key point for us. Output pixel: 10 m. Spatial context: the model uses attention over the whole input frame at coarse resolution, so a pixel's embedding is influenced by everything in its frame — and the supplement tells us how large the frame is: 1.28 km by 1.28 km, 128 pixels a side, for both training and the global inference run (which tiles the world in 960-m squares padded by 160 m on each side). So an AlphaEarth pixel never sees further than about 640 m in any direction. Time: the model takes a "valid period" — a start and end date — and produces a summary of everything observed in it. In principle any window; in practice the published product uses calendar years.

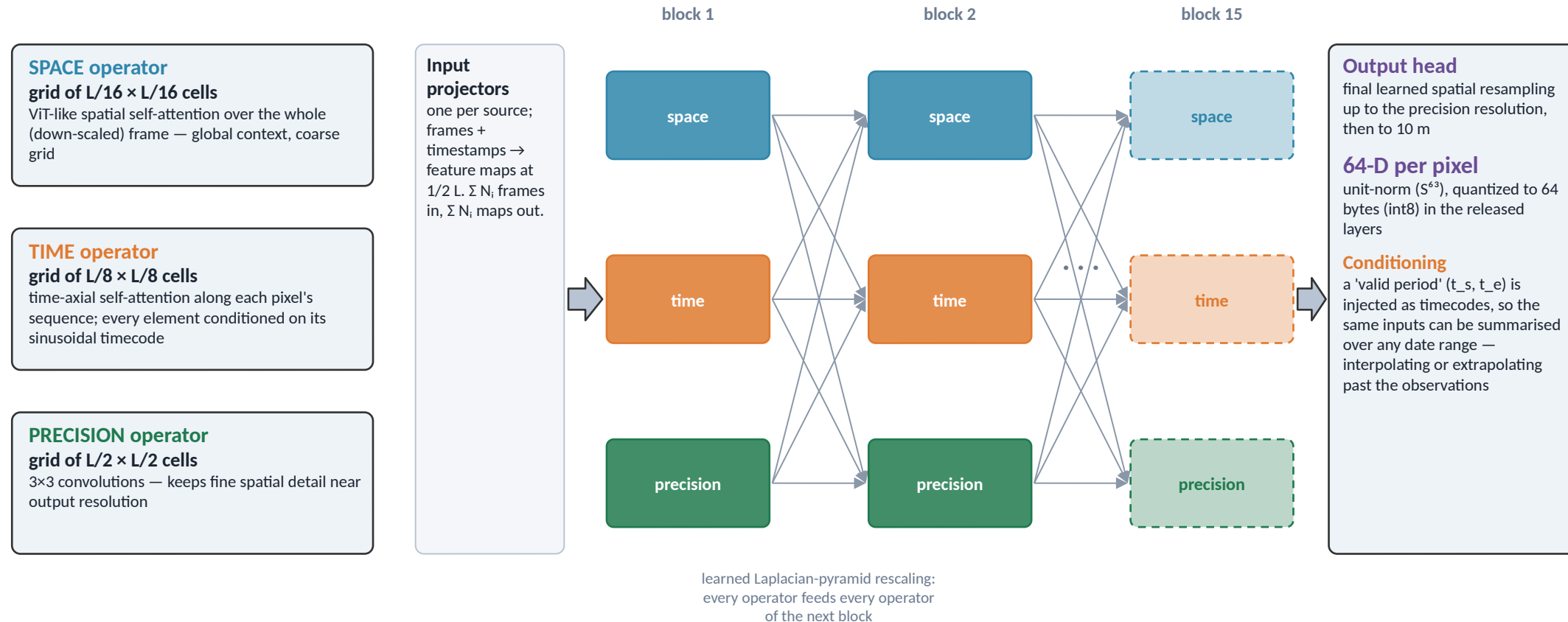
Output: 64 dimensions, normalised to length one (a "unit vector"), stored as 64 bytes. Coverage is land plus coastal and inland water; no open ocean.

Availability is the big caveat: only the dataset is released. The trained network is not, so nobody can run it on new data, new time windows, or the ocean.

"Error reduction ~23.9 %" means: across their 15 evaluation tasks, using AlphaEarth embeddings instead of the next-best method cut the error by about a quarter on average (about 10 % when only ten labels per class are available, about 4 % with one). The per-task numbers are published only as charts, so we do not quote them.

AlphaEarth — inside the STP encoder

L = side length of the square input frame, in pixels — $L = 128$ (1.28 km at 10 m) for training and inference (supplement S2.1, S8.2). Three operators at $L/16$, $L/8$ and $L/2$ exchange state through a learned Laplacian pyramid after every one of 15 blocks.



Training trio teacher video embedder with implicit decoders · student with identical architecture (consistency + batch-uniformity on the sphere) · text-alignment model with a frozen language model (CLIP-style). ~1 B and ~480 M variants trained; the 480 M one is deployed. **In the supplement (S2.4, S8.2):** 15 STP blocks; widths $D_S = 1024$ (space), $D_T = 512$ (time), $D_P = 128$ (precision); implicit decoders = 2-hidden-layer MLPs of width 512; vMF bottleneck $\kappa = 8 \cdot 10^3$; inference on 960 m tiles buffered to 1.28 km, outer 80 m trimmed. So the spatial receptive field is at most 1.28 km — the whole frame. **Still not stated:** the exact pyramid up/down-sampling kernels and how the three states are merged.

AlphaEarth — inside the STP encoder

This slide opens up the encoder. First the symbol: L is the side length of the square input frame, measured in pixels. The main text keeps it symbolic — "given a square input of L pixels a side" — but the supplement fixes it: frames are 128 by 128 pixels, 1.28 km at 10 m, so the three grids below are 8 by 8, 16 by 16 and 64 by 64 cells.

There are three "operators" — three different kinds of neural-network layer — that run in parallel inside every block:

The SPACE operator works on a coarse grid of $L/16$ by $L/16$ cells and uses self-attention, the mechanism from transformers in which every cell can look at every other cell. This gives global context across the frame at low resolution.

The TIME operator works on an $L/8$ grid and uses "time-axial" attention: each location looks only at its own history across all the dated frames. Every frame carries a sinusoidal "timecode" (a numeric encoding of its date), which is how the model knows when things happened.

The PRECISION operator works on the finest grid, $L/2$, with ordinary 3-by-3 convolutions to keep sharp spatial detail.

A "block" is one round of all three operators. After each block, a learned "Laplacian pyramid" rescaling step lets every operator hand its result to every operator of the next block: coarse results are up-sampled, fine results are down-sampled, so that information flows between scales. Repeat for 15 blocks (supplement S2.4, which also gives the widths: 1024 numbers per cell on the space path, 512 on the time path, 128 on the precision path).

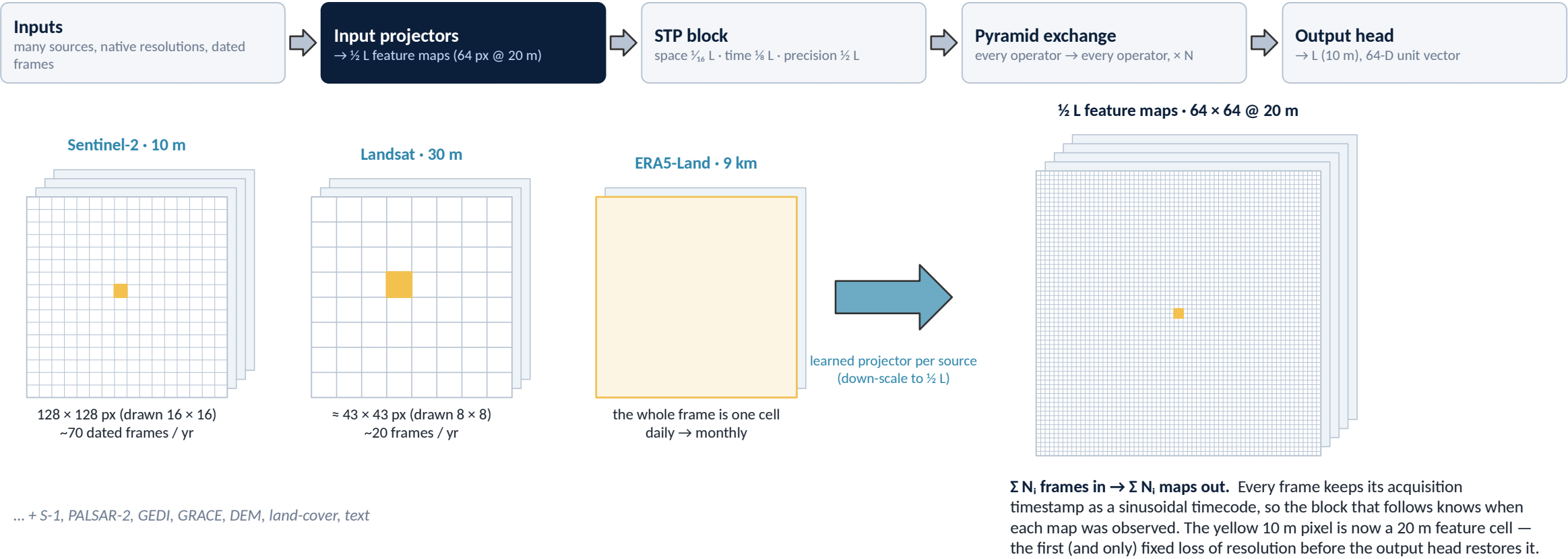
Input projectors bring each source's frames down to the $L/2$ grid first. The output head resamples back up to the precision resolution, then to 10 m, and produces the 64-number vector per pixel.

"Teacher/student" refers to the training recipe: one network (teacher) produces targets that a second, identical network (student) learns to match; a third model aligns embeddings with text descriptions in the style of CLIP.

Bottom line: because the space operator uses attention over the whole frame, the effective receptive field is the frame — 1.28 km, so at most about 640 m from the pixel in any direction. What the paper still does not spell out is the exact form of the pyramid resampling and how the three states are merged.

STP, step 1 of 4 — input projectors bring every source to $\frac{1}{2} L$

One frame of the same 1.28 km footprint from each sensor, at its own native resolution, becomes a 64×64 feature map at 20 m



... + S-1, PALSAR-2, GEDI, GRACE, DEM, land-cover, text

L = side length of the square input frame in pixels. $L = 128 \text{ px} @ 10 \text{ m}$ (1.28 km) is the paper's own frame (supplement S2.1, S8.2); the real depth is $N = 15$ blocks, we draw 4. Frame counts per year are typical values, not from the paper.

STP, step 1 of 4 — input projectors bring every source to $\frac{1}{2} L$

Steps 1 to 4 animate the previous slide with the paper's own frame: $L = 128$ pixels at 10 m, so the frame is 1.28 km across (supplement S2.1 and S8.2). Two things in these slides are simplified and the footers say so: we draw 4 blocks where the real network has 15, and the per-year frame counts are typical values rather than the paper's.

Step 1 is about the input projectors. Each data source arrives at its own native resolution over the same 1.28-km footprint: Sentinel-2 at 10 m gives a 128 by 128 grid (drawn coarser here); Landsat at 30 m gives about 43 by 43; ERA5-Land, a weather reanalysis at 9 km, gives just one value for the whole frame. Each source also arrives many times a year — Sentinel-2 roughly every five days, Landsat every 16.

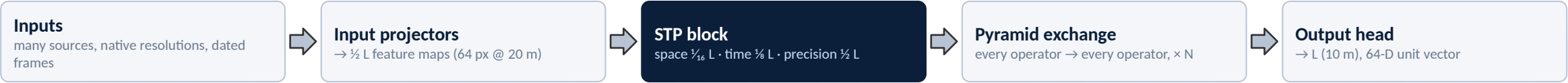
A learned projector per source converts each frame into a "feature map" — a grid of learned numbers rather than raw pixel values — at half the input resolution: 64 by 64 cells of 20 m. This is the only place where resolution is permanently reduced before the output head restores it.

The important detail is that each frame keeps its timestamp: it is carried along as a sinusoidal timecode so the time operator later knows when each observation was made. " $\sum N_i$ frames in, $\sum N_i$ maps out" simply means: however many frames come in from all sources combined, that many feature maps come out — nothing is merged yet.

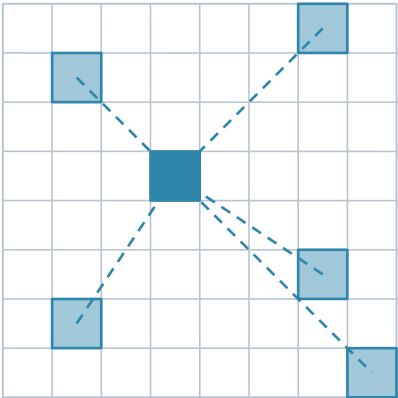
The yellow pixel is our 10-m output pixel; after this step it lives inside a 20-m feature cell.

STP, step 2 of 4 — three operators, three resolutions

Same 1.28 km footprint, three grids: what one cell can 'see' in each operator

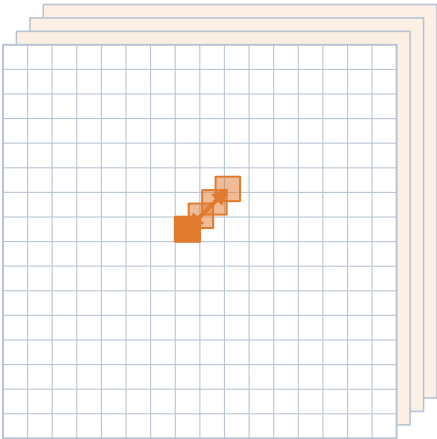


SPACE operator · $\frac{1}{16} L = 8 \times 8$ cells of 160 m



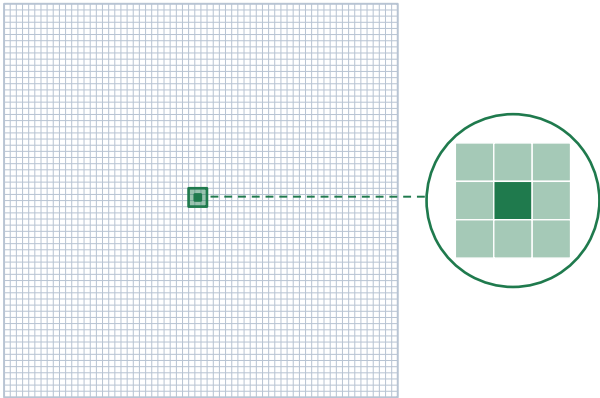
ViT-like self-attention: the cell attends to every other cell of the frame. Receptive field = whole frame, at coarse grain.

TIME operator · $\frac{1}{8} L = 16 \times 16$ cells of 80 m



Time-axial self-attention: the cell attends only to itself across all dated frames (stacked sheets), each tagged with its timecode.

PRECISION operator · $\frac{1}{2} L = 64 \times 64$ cells of 20 m



3 × 3 convolutions: the cell sees only its immediate neighbours (60 m), but at the finest grain the block carries.

Three views of one place, computed simultaneously. Space knows the whole frame coarsely, time knows the whole year at one spot, precision knows the fine local texture — none of them alone is the stencil; the exchange in step 3 is.

L = side length of the square input frame in pixels. L = 128 px @ 10 m (1.28 km) is the paper's own frame (supplement S2.1, S8.2); the real depth is N = 15 blocks, we draw 4. Frame counts per year are typical values, not from the paper.

STP, step 2 of 4 — three operators, three resolutions

Step 2 shows what one cell can "see" inside each of the three operators, on the same 1.28-km footprint.

SPACE (left): the grid has 8 by 8 cells of 160 m (that is $L/16$ with $L = 128$). The highlighted cell attends to every other cell — the dashed lines. So the space operator gives every cell a view of the whole frame, but only at coarse grain.

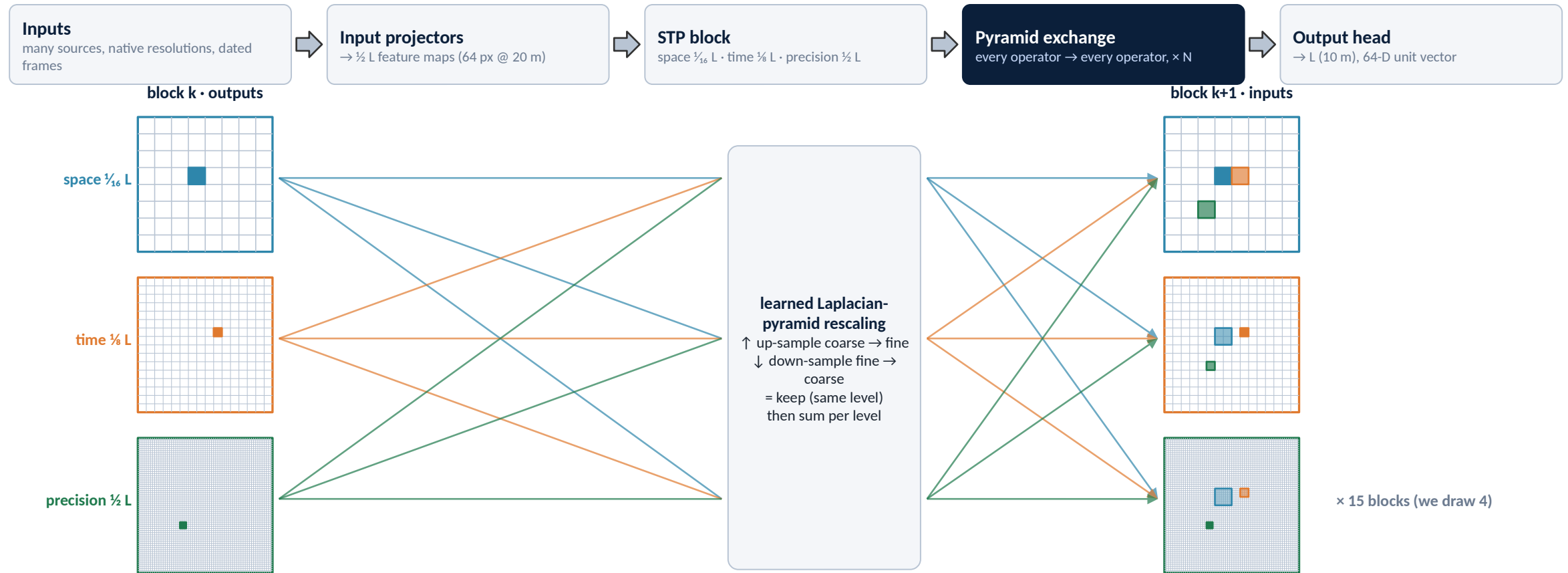
TIME (middle): 16 by 16 cells of 80 m. The stacked sheets are the dated frames. The highlighted cell attends only to itself across all the sheets — this is "time-axial" attention. It knows the whole history of one spot, but nothing about the neighbours.

PRECISION (right): 64 by 64 cells of 20 m. The operator is a 3-by-3 convolution, so a cell sees only its eight immediate neighbours — 60 m across — but at the finest grain the block carries. The magnifier shows that 3-by-3 neighbourhood.

The point of running three views at once: none alone is the stencil. Space knows the whole frame coarsely, time knows the whole year at one spot, precision knows the fine local texture. Step 3 is where they are combined.

STP, step 3 of 4 — the pyramid exchange between blocks

End of block k: each of the three maps is resampled to all three resolutions and summed into block k+1's inputs — repeat for N blocks



Follow the colours. The blue 160 m space cell arrives in the fine map as an 8×8 block of 20 m cells; the green 3×3 precision patch (60 m) arrives in the coarse map as a fraction of one 160 m cell; the orange time cell lands as $2 \times 2 / 4 \times 4$. After the sum, every operator in block k+1 starts from all three views. Repeat N times — the receptive field of the fine path grows to the whole frame while it keeps 20 m grain.

L = side length of the square input frame in pixels. *L* = 128 px @ 10 m (1.28 km) is the paper's own frame (supplement S2.1, S8.2); the real depth is *N* = 15 blocks, we draw 4. Frame counts per year are typical values, not from the paper.

STP, step 3 of 4 — the pyramid exchange between blocks

This is the sentence the walk-through was built to explain: "after each block, learned Laplacian-pyramid rescaling lets every operator feed every operator of the next block".

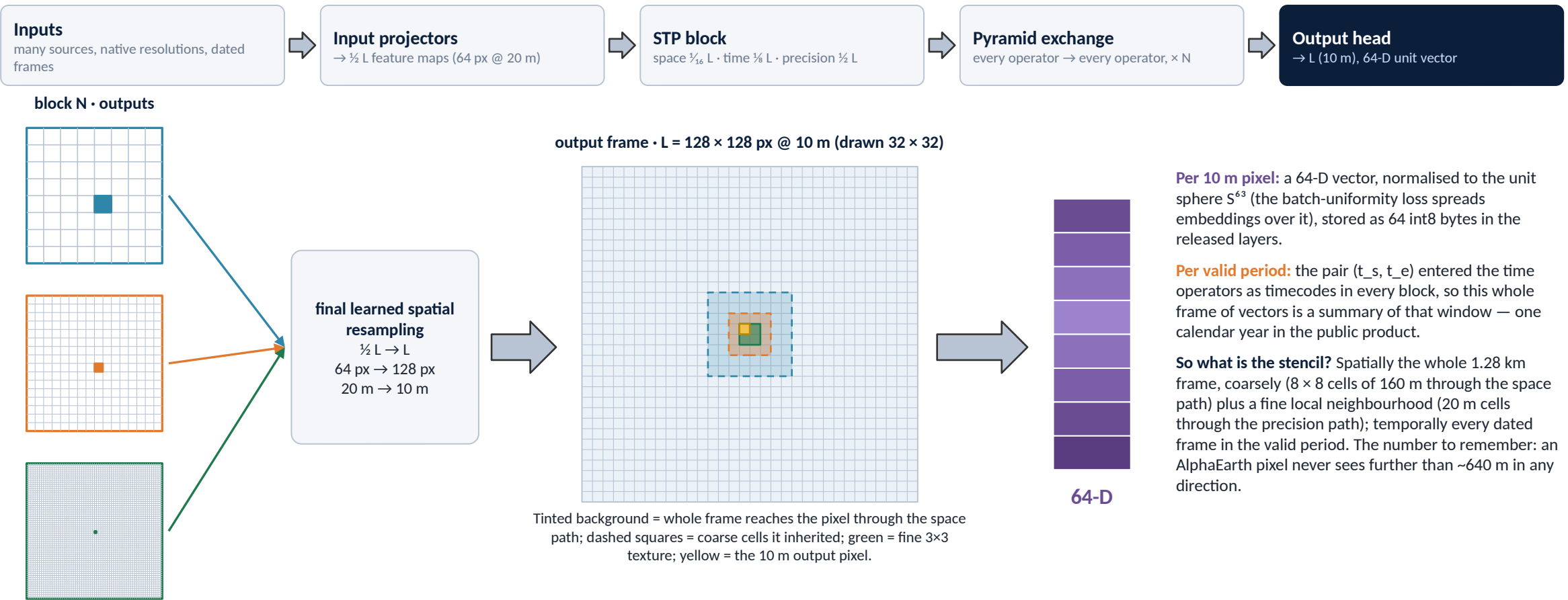
On the left are the three outputs of block k — the coarse space map, the medium time map, the fine precision map — each with one feature highlighted. In the middle is the rescaling stage. A Laplacian pyramid is a classic image-processing idea: an image is represented at several resolutions at once, and there are learned operations to move information up (coarse to fine, up-sampling) and down (fine to coarse, down-sampling). Here it is used to translate every map to every resolution.

On the right are the three inputs of block $k+1$. Each now contains all three colours: the blue space feature arrived in the fine map as an 8-by-8 block of 20-m cells, the green precision patch arrived in the coarse map as a fraction of one 160-m cell, the orange time cell landed as 2-by-2 or 4-by-4 blocks. Nine paths in total (three sources times three destinations), then a sum per level — the sum is our reading; the paper says "pass its state to each of the operators" without spelling out the merge.

Repeat 15 times (we draw 4). The consequence for our stencil question: after a few blocks, even the fine 20-m path has a receptive field that spans the whole frame, because it has received the space operator's global view several times — while keeping its own fine grain.

STP, step 4 of 4 — back to 10 m, 64-D per pixel

After block N: the $\frac{1}{2} L$ maps are resampled to the precision resolution and then to L; each output pixel is a unit vector on S^{63}



L = side length of the square input frame in pixels. L = 128 px @ 10 m (1.28 km) is the paper's own frame (supplement S2.1, S8.2); the real depth is N = 15 blocks, we draw 4. Frame counts per year are typical values, not from the paper.

STP, step 4 of 4 — back to 10 m, 64-D per pixel

Step 4 closes the loop. After the last block, the three maps are merged and a final learned spatial resampling takes the result from the L/2 grid (64 by 64 at 20 m) back to L (128 by 128 at 10 m).

The output frame in the middle shows, for one yellow 10-m pixel, what it has inherited: the tinted background stands for the whole frame reaching it through the space path; the dashed squares are the coarse cells it inherited from the space and time operators; the green patch is the fine local texture from the precision path.

Each output pixel becomes a 64-number vector. "Unit vector on S^{63} " is the mathematical way of saying the 64 numbers are scaled so their total length is exactly one — they sit on the surface of a 64-dimensional sphere. The batch-uniformity loss during training spreads the embeddings evenly over that sphere. In the released dataset each vector is stored as 64 bytes (one "int8", a whole number between -128 and 127, per dimension).

"Per valid period": the pair of dates (t_s , t_e) — start and end of the window — was fed into the time operators in every block as timecodes, so the whole frame of vectors is a summary of that window. In the public product the window is one calendar year.

So what is AlphaEarth's stencil? Spatially, the whole 1.28-km frame coarsely (8 by 8 cells of 160 m through the space path) plus a fine local neighbourhood (20-m cells through the precision path); temporally, every dated observation in the window. The number to remember: an AlphaEarth pixel never sees further than about 640 m.

TESSERA (University of Cambridge)

Pixel-wise 10 m annual embeddings from S-1 + S-2 time series — fully open (CC0), CVPR 2026

Who / when Cambridge Centre for Earth Observation; arXiv 2506.20380 (Jun 2025), CVPR 2026. Code github.com/ucam-eo/tessera, access via [geotessera](https://geotessera.org).

Inputs Sentinel-2 L2A (10 bands, 60 m bands dropped, resampled to 10 m) + Sentinel-1 RTC (VV, VH). Two modality-specific branches, fused late.

Architecture Per-pixel temporal transformer (4 layers) + GRU pooling per modality; encoder ~46 M params (1.34 B projector used only in training). Barlow Twins self-supervision on ~800 M pixel-series from 3,012 MGRS tiles; ~6,200 GPU-h on MI300X.

Stencil Strictly one 10 m pixel — no spatial context at all. One calendar year; 40 valid dates sampled per modality with learnable day-of-year embeddings (irregular cadence tolerated).

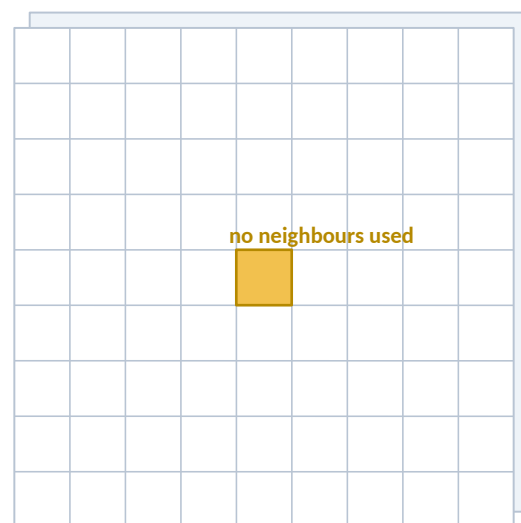
Output 128-D per pixel-year, int8 (QAT). Years 2017–2025, global land (water masked); tile-sampled coverage, gaps on request.

Availability Weights (CC0), code (MIT), precomputed embeddings via GeoTessera CDN. Self-run: ~10 h per 110 km tile-year (128-core CPU + A30).

Results Frozen + MLP head: PASTIS-R mIoU 50.7, TreeSatAI-TS F1 78.0, canopy height RMSE 12.2 m; holds up until <~20 valid S-2 obs/yr.

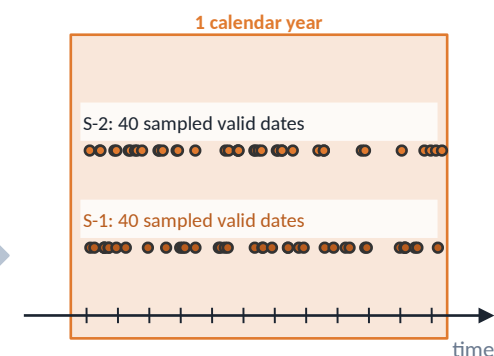
Stencil — what feeds one embedding

SPACE



Output pixel: 10 m × 10 m.
Receptive field = the same single pixel. Neighbouring pixels never enter.
Two modality sheets: S-2 (10 bands) and S-1 (2 bands).

TIME



Fixed window; dates irregular, encoded by day-of-year.
Sampling with replacement pads pixel-years with <40 clear observations.

OUTPUT



128-D
per 10 m pixel per year
int8

TESSERA (University of Cambridge)

TESSERA comes from the University of Cambridge (published June 2025, accepted at CVPR 2026) and is the most radical design in the deck: it uses no spatial context at all.

Inputs: Sentinel-2 optical (10 bands) and Sentinel-1 radar (2 polarisations, VV and VH). Sentinel-1 is radar, so it sees through clouds; Sentinel-2 is optical.

Architecture: for each 10-m pixel separately, a small transformer reads the pixel's own time series — 40 dates sampled from one calendar year for each sensor — and a GRU (a type of recurrent network) pools them into one vector. The two sensors have their own branches which are fused at the end. The encoder is only about 46 million parameters; a much larger "projector" is used only during training and then discarded. Training uses Barlow Twins, a self-supervised objective that asks two differently-sampled views of the same pixel-year to produce the same embedding while keeping the embedding's dimensions non-redundant — no reconstruction of inputs is involved.

Stencil: strictly one 10-m pixel; neighbours never enter. One calendar year; dates are irregular, and the model is told the day of year of each observation. If a pixel has fewer than 40 clear observations, dates are sampled with replacement to fill the 40 slots.

Output: 128 numbers per pixel per year, stored as int8. Years 2017–2025, land only (water masked). Everything is released: weights (CC0, the most permissive licence), code (MIT), and precomputed embeddings through the GeoTessera service.

Results quoted are with the embeddings frozen and only a small head trained on top: PASTIS-R crop segmentation mIoU 50.7, TreeSatAI tree-species F1 78.0, canopy height error 12.2 m. mIoU is "mean intersection over union", the standard segmentation score (100 = perfect overlap). F1 is the balanced average of precision and recall for classification.

Why it matters for us: TESSERA proves that a per-pixel temporal encoder with zero spatial context is competitive on land. That is the opposite extreme from our 3-by-3 patch, and a useful reference point.

OlmoEarth (Ai2)

Open multimodal space-time ViT family (Nano→Large); embeddings computed on demand, not precomputed

Who / when Allen Institute for AI (+ UW, ASU, UBC). v1 Nov 2025 (arXiv 2511.13655), v1.1 May 2026, v1.2 later 2026. Weights on Hugging Face; custom 'OlmoEarth Artifact License' (no military/extractive use).

Inputs Sentinel-1, Sentinel-2, Landsat-8 monthly time series, all resampled to 10 m; static layers (OSM, WorldCover, CDL, SRTM, canopy height, WorldCereal) as inputs/targets.

Architecture ViT over space × time × band-group tokens (FlexiViT patches 1–8 px in v1; single-bandset token merging in v1.1, RoPE in v1.2). 'Latent MIM Lite': masked modelling with patch-discrimination + instance-contrastive losses. Sizes: Nano 1.4 M · Tiny 6.2 M · Base 89 M · Large 308 M (encoder).

Stencil Tile 2.56 km × 2.56 km = 256 × 256 px @ 10 m; token = 1–8 px patch (≤ 80 m). Up to 12 monthly steps over one year, missing months masked.

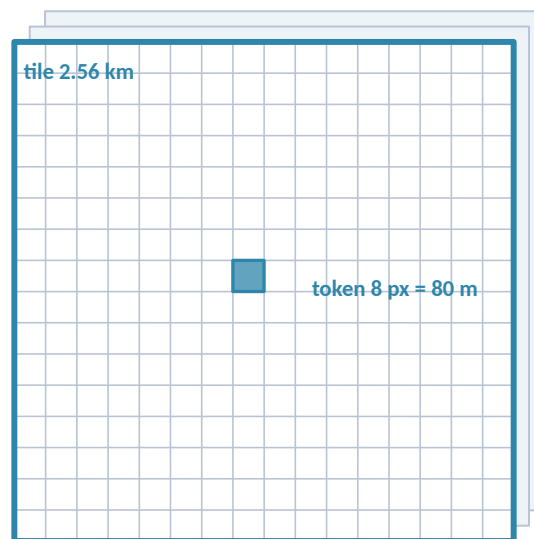
Output Per-patch, per-timestep tokens (Base 768-D, Large 1024-D) + pooled vector. No standing global product; Studio exports embeddings for any AOI at 10–80 m, 1–12 monthly steps (int8 COGs).

Results Best on 15/24 tasks (kNN/linear) and 19/29 (fine-tuned) vs. Prithvi v2, TerraMind, Galileo, DINOv3; matches/exceeds AlphaEarth after fine-tuning on 5 tasks.

Pretraining data 285 k samples × 2.56 km × 1 yr, stratified over 120 OSM classes (~10 TB) — land/infrastructure-biased.

Stencil — what feeds one embedding

SPACE

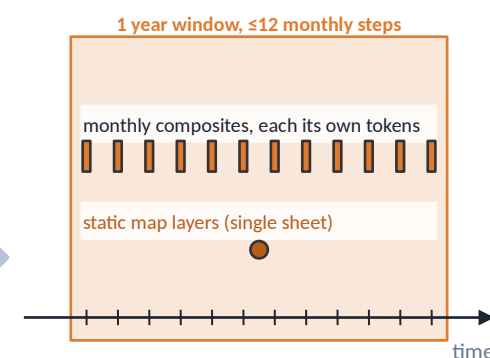


Tile: 256 × 256 px @ 10 m = 2.56 km (drawn 16 × 16 for legibility).

Token: flexible 1–8 px patch → 10–80 m per token, full attention across the tile.

Sheets: S-1, S-2, Landsat + static maps.

TIME



Each month is tokenised separately; attention runs across space, time and modality. Temporal masking (v1.2) trains the model to fill missing months.

OUTPUT



768-D

per patch per month (Base) + pooled tile vector

OlmoEarth (Ai2)

OlmoEarth is the Allen Institute for AI's family of open models, first released November 2025; version 1.1 (May 2026) merged Sentinel-2's band groups into a single token to cut compute, and version 1.2 added rotary position encodings. It is a "space-time ViT": a Vision Transformer that tokenises not just space but also time and sensor band groups.

Inputs: Sentinel-1, Sentinel-2 and Landsat-8 monthly time series, all resampled to 10 m, plus static map layers (OpenStreetMap, land cover, crop data, elevation, canopy height) used as extra inputs and as prediction targets.

Architecture: the tile is 2.56 km square (256 by 256 pixels at 10 m). Tokens are flexible patches of 1 to 8 pixels — so 10 to 80 m — and up to 12 monthly steps are tokenised separately. Attention runs across space, time and modality together. The training objective, "Latent MIM Lite", is masked modelling: hide some tokens, predict them in the embedding space rather than as pixels, plus contrastive terms. Four sizes: Nano (1.4 M parameters), Tiny (6 M), Base (89 M), Large (308 M).

Output: one vector per patch per month (768 dimensions for Base, 1024 for Large) plus a pooled vector for the tile. There is no global precomputed product; instead the "OlmoEarth Studio" computes embeddings on demand for any region you draw.

Results: in the authors' own benchmark the model is best on 15 of 24 tasks with frozen embeddings and 19 of 29 when fine-tuned; on five tasks a fine-tuned OlmoEarth matches or beats AlphaEarth.

Licence: open weights, but under a custom licence that forbids some uses (military, extractive industries).

For us: OlmoEarth is the closest analogue to a general space-time transformer we could actually retrain — open weights, monthly tokens, a masked latent objective — but its pretraining sample is land-biased.

TerraMind (IBM Research + ESA Φ-lab)

Any-to-any generative multimodal EO model; 'Thinking in Modalities' — ICCV 2025

Who / when IBM Research Europe, ESA Φ-lab, ETH Zürich, FZ Jülich, NASA IMPACT, Univ. of Iceland. arXiv 2504.11171 (Apr 2025); TerraMesh dataset 2504.11172. Apache 2.0, weights on Hugging Face (ibm-esa-geospatial), TerraTorch integration. Tiny/Small edge variants Jun-Jul 2025.

Inputs Pixel-level: S-2 L2A & L1C, S-1 GRD & RTC, DEM, RGB, NDVI. Token-level: LULC classes, captions, geolocation. All co-registered at 10 m in 224 px tiles.

Architecture Dual-scale encoder-decoder: token path (cross-modal correlation) + pixel path (fine detail). Per-modality FSQ-VAE tokenizers (16 k codebook). Objective: masked cross-modal token reconstruction over 500 B tokens from ~9 M TerraMesh samples. Base: 6 d, Large: 10 d on 32 A100s.

Stencil Tile 224×224 px @ 10 m = 2.24 km; token 16×16 px = 160 m → $14 \times 14 = 196$ tokens per modality. Single co-registered timestamp per sample — no multi-temporal fusion.

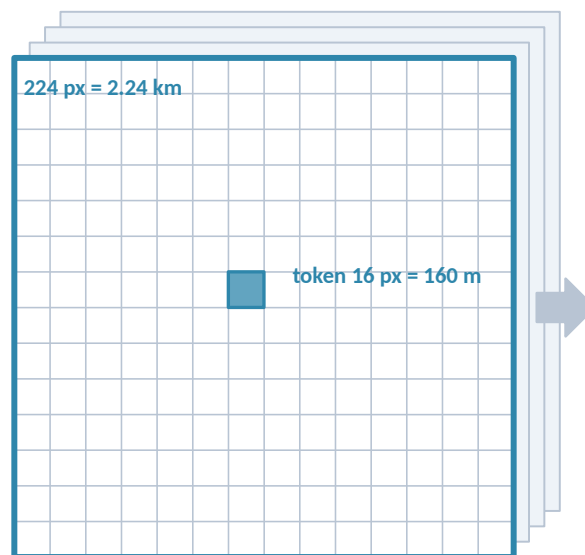
Output 196 tokens per modality of 768-D (Base) / 1024-D (Large), mergeable to one vector; plus generated rasters of missing modalities (TiM). No precomputed embedding archive.

Results PANGAEA avg mIoU: Large 59.6, Base 58.4 (≥ 3 pp over CROMA, DOFA and the other GeoFMs in that table); only GeoFM beating task-specific U-Nets across the benchmark.

Ocean TerraMesh is land/ecoregion-stratified; no marine modalities. Open-ocean behaviour untested.

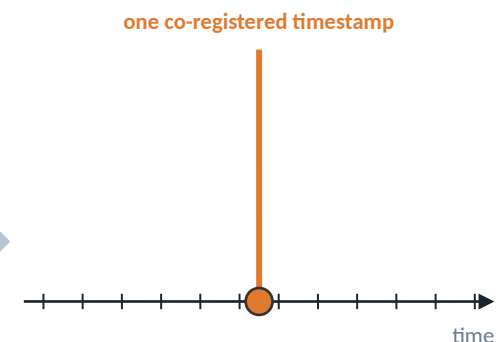
Stencil — what feeds one embedding

SPACE



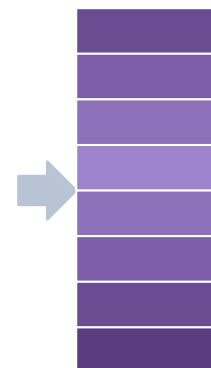
Tile: 224×224 px @ 10 m = 2.24 km × 2.24 km.
Token: 16×16 px = 160 m; 14×14 grid per modality.
Up to ~9 modality sheets, each with its own tokenizer.

TIME



All modalities come from the same date and place.
A sample is a snapshot: every modality is aligned to one acquisition.
Temporal change is not part of the pretraining stencil.

OUTPUT



768-D

per 160 m token, per modality
(Base; Large 1024-D)
or merged per tile

TerraMind (IBM Research + ESA Φ-lab)

TerraMind is a joint IBM Research and ESA Φ-lab model (April 2025, ICCV 2025). Its distinguishing feature is that it is generative and "any-to-any": given some modalities it can produce the missing ones.

Inputs come in two kinds. Pixel-level: Sentinel-2 optical, Sentinel-1 radar, elevation, RGB, vegetation index (NDVI). Token-level: land-use class maps, text captions, geographic coordinates. All are aligned on 10-m, 224-pixel tiles.

Architecture: each modality has its own "tokenizer" — a small network that turns its raster into discrete codes from a 16,000-entry codebook (an FSQ-VAE, a kind of quantising autoencoder). The main encoder-decoder then learns to predict masked codes of one modality from the others. "Thinking in Modalities" (TiM) is the trick of generating a missing modality — say land cover from optical — and feeding it back in as extra input. Trained on 500 billion tokens from about 9 million TerraMesh samples.

Stencil: tile 2.24 km (224 pixels at 10 m); token 16 pixels = 160 m, so 14 by 14 tokens per modality. One co-registered timestamp per sample — every modality comes from the same date and place; there is no multi-temporal fusion in pretraining.

Output: 196 tokens per modality, 768 dimensions for the Base model and 1024 for Large, or one merged vector per tile, plus generated rasters. No precomputed embedding archive; weights are Apache 2.0 (fully open) on Hugging Face.

Results: on the PANGAEA benchmark (a standard suite of segmentation tasks) TerraMind Large scores 59.6 mean IoU, the only foundation model to beat U-Nets trained from scratch on that suite.

Ocean: the training data is stratified by land ecoregion; there are no marine modalities and open-ocean behaviour is untested.

Prithvi-EO 2.0 (IBM + NASA)

Multi-temporal HLS masked autoencoder at 30 m; 300 M and 600 M, with temporal-location (TL) variants

Who / when IBM Research + NASA IMPACT, trained at JSC; arXiv 2412.02732 (Dec 2024, v3 Mar 2026). Weights on Hugging Face (ibm-nasa-geospatial); fine-tuning via TerraTorch. Succeeds Prithvi-EO 1.0 (100 M).

Inputs HLS v2 (Harmonized Landsat-Sentinel-2) at 30 m, 6 bands (B02–B07: blue, green, red, NIR-narrow, SWIR1, SWIR2). 4.2 M training samples, 2014–2023; sea-only tiles removed with Fmask, Greenland excluded.

Architecture ViT-L (300 M: 1024-D, 24 layers) / ViT-H (600 M: 1280-D, 32 layers) masked autoencoder, mask ratio 0.75, 3D sin-cos positions. TL variants add lat/lon + date encodings with metadata dropout.

Stencil Tile 224×224 px @ 30 m = 6.72 km. Token 16×16 px = 480 m (300 M) or 14×14 px = 420 m (600 M); tubelet depth 1, so each of 4 timestamps gets its own token grid. Frames 1–6 months apart (seasonal), not a fixed window; single-frame inference supported.

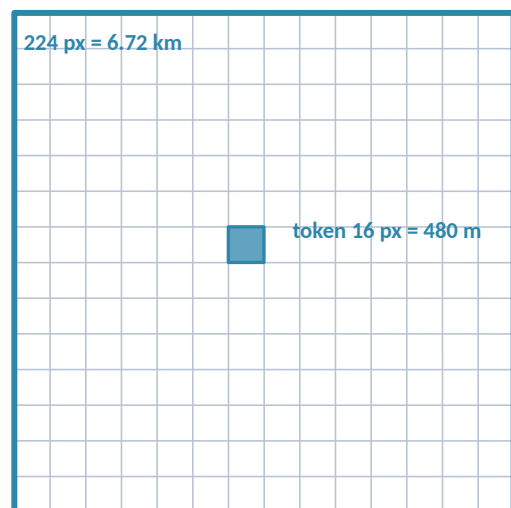
Output 4×196 (or 4×256) tokens of 1024/1280-D. No precomputed embedding product; typical use is full fine-tuning of encoder + light decoder.

Results GEO-Bench: 600 M-TL best on average, ~8 % over v1.0 and ahead of six other GFMs incl. DOFA and Scale-MAE (per-set numbers are charts); tasks: floods, burn scars, crops, landslides, PASTIS, Sen4Map.

In orbit May 2026: deployed on Kanyini satellite and ISS IMAGIN-e for onboard flood/cloud detection (still the 2.0 architecture).

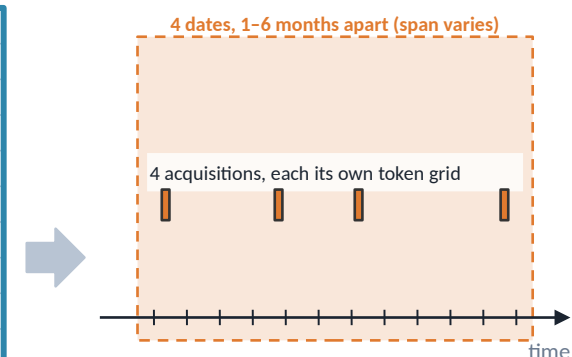
Stencil — what feeds one embedding

SPACE



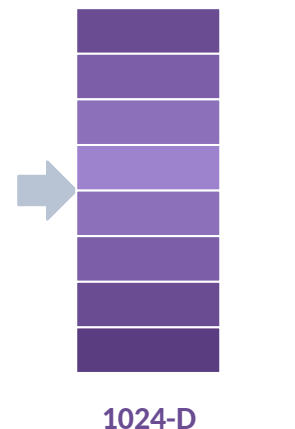
Tile: 224×224 px @ 30 m = 6.72 km $\times$ 6.72 km.
Token: 16×16 px = 480 m (300 M) $\cdot$ 14×14 px = 420 m (600 M).
One sheet: 6 HLS bands.

TIME



Time is a stack of 4 snapshots with 3D position codes — not a period summary. TL variants inject acquisition date + lat/lon as extra encodings.

OUTPUT



per 480 m token, per date
300 M: 1024-D $\cdot$ 600 M: 1280-D

Prithvi-EO 2.0 (IBM + NASA)

Prithvi-EO 2.0 is IBM and NASA's second-generation Earth observation model (December 2024). It matters here for two reasons: it is a widely used open baseline, and IBM's ocean model on the next slide reuses its recipe.

Inputs: HLS, the Harmonized Landsat-Sentinel-2 product, at 30 m with six spectral bands (blue, green, red, near-infrared, and two shortwave-infrared bands). 4.2 million training samples from 2014–2023. Importantly for us, sea-only tiles were explicitly removed using a cloud/water mask, and Greenland was excluded.

Architecture: a masked autoencoder (MAE) — the network sees a tile with 75 % of its patches hidden and learns to reconstruct them. Two sizes: 300 million parameters (a "ViT-Large", 1024-dimensional embeddings, 24 layers) and 600 million (ViT-Huge, 1280 dimensions, 32 layers). The "TL" variants additionally receive the acquisition date and latitude/longitude as extra encodings.

Stencil: tile 224 pixels at 30 m = 6.72 km. Token 16 pixels = 480 m for the 300 M model, 14 pixels = 420 m for the 600 M model (we checked the published configuration files). Time: four frames, each 1–6 months apart to capture seasons; the "tubelet depth" is 1, meaning each frame gets its own grid of tokens rather than frames being merged into one token — time is fused only inside attention. Single-frame use is also supported.

Output: one vector per token per frame; no precomputed product. The typical way to use it is full fine-tuning — retraining the encoder together with a small decoder for your task — through IBM's TerraTorch toolkit.

Results: on GEO-Bench (a 12-task standard benchmark) the 600 M-TL model is the best on average, about 8 % above version 1.0 and ahead of six other foundation models; the per-dataset scores are published as charts, so we quote no single number. In May 2026 the model was deployed on a satellite for onboard flood and cloud detection.

Granite-Geospatial-Ocean (IBM · STFC · PML · Exeter)

The Prithvi-EO recipe moved onto Sentinel-3 ocean colour + SST — github.com/ibm-granite/geospatial

Who / when IBM Research Europe (UK), STFC Hartree Centre, Plymouth Marine Lab, Univ. of Exeter (UK HNCID). Released 6 Oct 2025; arXiv 2509.21273 'A Sentinel-3 foundation model for ocean colour'. Apache 2.0; weights on Hugging Face, code + TerraTorch notebook at github.com/ibm-granite/geospatial.

Inputs Sentinel-3 OLCI Level-2 reflectance, 16 bands (Oa01–Oa12, Oa16–18, Oa21) + SLSTR sea-surface temperature = 17 channels at 300 m. ~512 k tiles (the paper also gives 470 k train + 50 k val), 2017–2021, balanced over 83 Longhurst provinces and months.

Architecture Prithvi-EO ViT-MAE with the tile shrunk for the coarser sensor: embed 512-D, 12 layers, 8 heads, decoder 256-D × 4; ~50 M params (larger gave no gain). RMSE reconstruction loss on masked patches.

Stencil Tile 42 × 42 px @ 300 m = 12.6 km × 12.6 km; token 2 × 2 px = 600 m → 21 × 21 = 441 tokens. num_frames = 1: a single acquisition, no temporal stacking (fine-tuning pairs each in-situ sample with cloud-free imagery from a 6-day window centred on it).

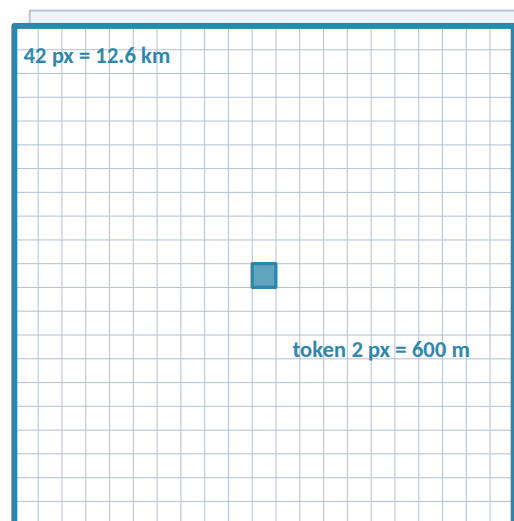
Output 441 × 512-D tokens per tile; downstream pixel-wise regression heads (TerraTorch) for chlorophyll-a and primary production.

Results Chl-a (188 in-situ points): RMSE 0.14 ± 0.05 vs 0.16 ± 0.10 log₁₀ mg m⁻³ from scratch; primary production (103 points): 0.39 ± 0.04 vs 0.42 ± 0.07 log₁₀ mgC m⁻² d⁻¹. Gains grow as labels shrink (useful at 25 % of labels).

Caveats Surface-only, cloud-limited optical; single snapshot; authors say not yet competitive with operational algorithms; under-predicts bloom-level chl-a.

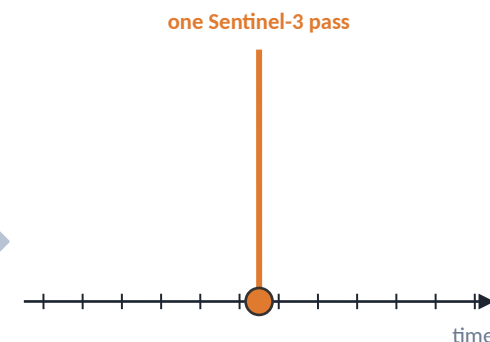
Stencil — what feeds one embedding

SPACE



Tile: 42 × 42 px @ 300 m = 12.6 km × 12.6 km (21 × 21 tokens drawn to scale).
Token: 2 × 2 px = 600 m.
Two sheets: 16 OLCI bands + 1 SST band.

TIME



num_frames = 1 — no temporal stacking. Single-timestep stencil: the 4-frame Prithvi option is switched off. Fine-tuning pairs each in-situ measurement with imagery from a 6-day window centred on it.

OUTPUT



512-D

per 600 m token
per single acquisition

Granite-Geospatial-Ocean (IBM • STFC • PML • Exeter)

This is IBM's Prithvi-based ocean model, developed with the UK's STFC Hartree Centre, Plymouth Marine Laboratory and the University of Exeter, released October 2025. The code lives in the `ibm-granite` GitHub organisation; we read its configuration files directly to get the numbers below.

Inputs: Sentinel-3 OLCI, the ocean-colour instrument, 16 spectral bands at 300 m, plus sea-surface temperature from the SLSTR instrument — 17 channels in total. About 512,000 tiles from 2017–2021 (the paper also quotes 470,000 for training plus 50,000 for validation), sampled evenly across 83 "Longhurst provinces" (a standard division of the ocean into biogeochemical regions) and across months.

Architecture: the Prithvi masked-autoencoder recipe with the network shrunk to fit the coarser sensor: embedding size 512, 12 layers, about 50 million parameters, trained to reconstruct masked patches under a root-mean-square-error loss. The authors report that larger networks gave no gain.

Stencil: tile 42 by 42 pixels at 300 m = 12.6 km square. Token 2 by 2 pixels = 600 m, so 21 by 21 = 441 tokens. `num_frames = 1`: a single acquisition, no temporal stacking — the four-frame option of Prithvi is switched off. For fine-tuning, each ship measurement is paired with cloud-free imagery from a six-day window centred on it.

Output: 441 vectors of 512 dimensions per tile. Downstream, a small regression head is trained to predict chlorophyll-a (a proxy for phytoplankton) and primary production (how much carbon the phytoplankton fix).

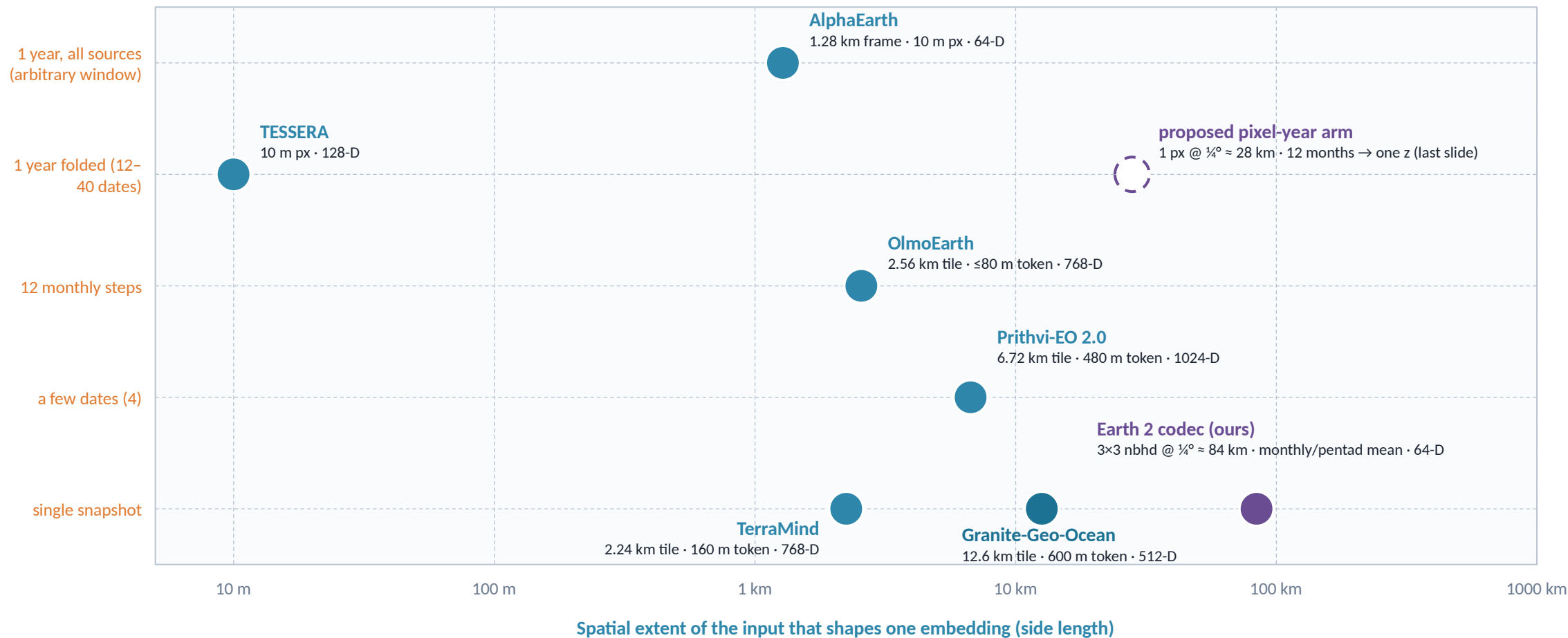
Results: the numbers are RMSE — root-mean-square error — in log10 units, because these quantities vary over orders of magnitude. With the foundation model: 0.14 for chlorophyll, versus 0.16 training the same network from scratch; 0.39 versus 0.42 for primary production. The gains are small and within the fold-to-fold spread; the authors' stronger claim is that the advantage grows when labels are scarce.

Caveats: surface-only, optical (so cloud-limited), one snapshot in time. It is a model of ocean colour — biology at the surface — not of the ocean's physical state.

Stencils side by side

Spatial footprint that feeds one embedding (log scale) vs. how much time it summarises

Temporal window per embedding



Earth 2 shown for reference: the codec's channel token sees a 3 × 3 neighbourhood of 1/4° (or 1°) cells at one monthly or 5-day mean — a snapshot stencil ~7× wider than the widest EO tile here (Granite, 12.6 km).

Stencils side by side

This chart puts all six models — and our own — on two axes that summarise the whole deck.

Horizontal axis (logarithmic): the spatial extent of the input that shapes one embedding, from 10 m to 1000 km. Vertical axis (categorical): how much time one embedding summarises, from a single snapshot at the bottom to a full year of all sources at the top.

Reading the points: TESSERA sits at 10 m on the far left — a per-pixel output with no spatial context at all — and AlphaEarth at 1.28 km, the frame each of its 10-m pixels is computed from; both sit at the top because both fold a whole year into one vector. TerraMind and Granite are single snapshots at 2–13 km tile scale. Prithvi stacks four dates; OlmoEarth stacks twelve monthly steps.

Earth 2, our codec, is the purple dot at about 84 km: each channel token sees a 3-by-3 neighbourhood of quarter-degree cells (a quarter degree is about 28 km) at one monthly or five-day mean. It is a snapshot stencil roughly seven times wider than the widest Earth-observation tile here (Granite, 12.6 km). The hollow dot is the proposed "pixel-year" experiment from a later slide, which would move us up to the "one year folded" row.

Two design axes separate all these models: per-pixel versus patch, and snapshot versus period summary. Every later recommendation in the deck comes back to these two axes.

Summary I — what goes into each embedding

Inputs, spatial stencil, temporal stencil, output vector

Model	Inputs (sensors / modalities)	Spatial stencil	Temporal stencil	Output per embedding
AlphaEarth (DeepMind)	S-2, Landsat 8/9, S-1, PALSAR-2, GEDI, ERA5-Land, GRACE, DEM, NLCD, text	10 m output pixel inside a 1.28 km (128 px) frame; attention global within the frame	Period summary; arbitrary valid window in the model, calendar year in the product	64-D unit vector, int8; per pixel per year
TESSERA (Cambridge)	S-2 L2A (10 bands) + S-1 RTC (VV, VH)	Single 10 m pixel — no spatial context	1 calendar year; 40 sampled dates per modality, day-of-year encoded	128-D int8; per pixel per year
OlmoEarth (Ai2)	S-1, S-2, Landsat-8 monthly + static maps (OSM, WorldCover, CDL, SRTM, canopy, WorldCereal)	2.56 km tile (256 px @ 10 m); 1–8 px tokens (10–80 m)	≤12 monthly steps over 1 year, each step tokenised; temporal masking	768-D (Base) / 1024-D (Large) per patch per step + pooled vector
TerraMind (IBM/ESA)	S-2 L2A/L1C, S-1 GRD/RTC, DEM, RGB, NDVI, LULC, captions, geolocation	2.24 km tile (224 px @ 10 m); 16 px = 160 m tokens, 14×14 grid	Single co-registered timestamp	196 × 768-D tokens per modality; merged 768-D; generated modalities
Prithvi-EO 2.0 (IBM/NASA)	HLS v2 at 30 m, 6 bands (blue, green, red, NIR, SWIR1, SWIR2)	6.72 km tile (224 px @ 30 m); 480 m (300 M) or 420 m (600 M) tokens	4 dated frames, 1–6 months apart, tubelet depth 1; single frame allowed	1024-D (300 M) / 1280-D (600 M) per token per frame
Granite-Geo-Ocean (IBM/STFC/PML)	Sentinel-3 OLCI L2, 16 bands + SLSTR SST (17 ch) at 300 m	12.6 km tile (42 px @ 300 m); 2 px = 600 m tokens, 21×21 grid	Single acquisition (num_frames = 1)	512-D per token; regression heads for chl-a, primary production

Summary I — what goes into each embedding

A reference table of what goes into each embedding, condensing the model slides. Column by column:

Inputs: the sensors and data sources. S-1 and S-2 are Sentinel-1 (radar) and Sentinel-2 (optical); HLS is the harmonised Landsat–Sentinel product; OLCI and SLSTR are the Sentinel-3 ocean-colour and temperature instruments.

Spatial stencil: what area feeds one embedding — the output pixel size, the token size (the patch the network reasons about) and the tile size (the block it processes at once).

Temporal stencil: whether the model summarises a period (AlphaEarth, TESSERA), stacks dated frames (OlmoEarth, Prithvi), or takes a single snapshot (TerraMind, Granite).

Output: the vector dimension and what one vector describes — a pixel-year, a token, a tile.

The table is designed to be read across a row to reconstruct a model's stencil, or down a column to compare one design choice.

Summary II — what each project hands you

Weights, precomputed embeddings, licence, ocean coverage, and the intended way to use it

Model	Open weights / code	Precomputed embeddings	Licence	Ocean?	Intended use / evaluation style
AlphaEarth	No — model and code unreleased	Yes — global 10 m annual 2017–2024 in Earth Engine + GCS	CC-BY 4.0 (data)	Coastal/inland water only	Frozen embeddings + linear/shallow heads on sparse labels; 'featurization' replacement
TESSERA	Yes — weights CC0, code MIT, full pipeline	Yes — 10 m annual 2017–2025 via GeoTessera (tile-sampled)	CC0 / MIT	Land only (water masked)	Frozen per-pixel embeddings + MLP/conv heads; label-efficient mapping
OlmoEarth	Yes — 4 sizes on Hugging Face	On demand — Studio exports for any AOI; no global archive	OlmoEarth Artifact License (use restrictions)	Not in pretraining sample	Both: kNN/linear probes and full fine-tuning; platform for NGOs/agencies
TerraMind	Yes — tiny/small/base/large + tokenizers	No	Apache 2.0	Land-stratified TerraMesh	Fine-tuning via TerraTorch; any-to-any generation; edge/onboard variants
Prithvi-EO 2.0	Yes — 300 M / 600 M, ±TL	No	Apache 2.0 (HF) / MIT (GitHub) — verify	Sea-only tiles removed	Full fine-tuning of encoder + decoder (TerraTorch); GEO-Bench
Granite-Geo-Ocean	Yes — HF weights + GitHub notebook	No	Apache 2.0	Yes — open ocean, surface optical + SST	Fine-tune regression heads on ~100–200 in-situ points; ocean-colour biogeochemistry

Summary II — what each project hands you

The companion table: what each project actually hands you, which is often as important as the architecture.

Open weights: whether the trained network is downloadable, so you can run it on your own data or fine-tune it. AlphaEarth is the exception — no weights, no code.

Precomputed embeddings: whether a ready-made global product exists. AlphaEarth and TESSERA yes; OlmoEarth on demand through its Studio; the others no.

Licence: CC-BY and CC0 are Creative Commons licences (attribution required, or no restrictions); Apache 2.0 and MIT are permissive software licences; OlmoEarth's custom licence has use restrictions. Prithvi's licence is stated differently on Hugging Face and GitHub, hence "verify".

Ocean: green means open ocean is in the training data, red means it is masked out or not targeted. Only Granite is green among the six.

Intended use: "frozen embeddings" means you keep the network fixed and train only a small head on top; "fine-tuning" means you retrain the network for your task; kNN is k-nearest-neighbours, the simplest possible classifier, used to test how good raw embeddings are.

Who embeds the ocean — and how deep?

Representation models whose pretraining contains sea pixels: the published ones stop at the surface; Earth 2 is the one with the interior (checked Aug 2026)

Model (org, year)	Ocean data in pretraining	Open ocean? depth?	Stencil	Embedding / availability	What it is for
Earth 2 codec (ours, 2026)	Gridded ocean state at $\frac{1}{4}^\circ$ / 1° : T and S at depth from Argo (plus pre-Argo gaps marked 'never measured'), SSH, currents, wind and surface forcing; 14–25 channels, monthly or pentad means	Yes — open ocean, surface to ~2000 dbar interior	3×3 patch per channel token; one pixel-month snapshot; 24-month history in stage 2 with a 222–4444 km stencil	64-D per pixel-month (128-D variant training); weights + embeddings ours to release	Ocean-state field prediction; frozen-embedding probes to AMOC transport (RAPID, MOVE)
Granite-Geospatial-Ocean (IBM • STFC • PML • Exeter, 2025)	Sentinel-3 OLCI 16 bands + SLSTR SST, 300 m; 512 k tiles 2017–2021, balanced over 83 Longhurst provinces	Yes — surface optical only	42 px = 12.6 km tile; 600 m tokens; single acquisition	512-D per token; Apache 2.0 weights on HF + TerraTorch notebook	Ocean-colour biogeochemistry: chl-a, primary production (surface optical)
OceanSAR-1 / -2 (Galeio, Apr 2025 / Jan 2026)	Sentinel-1 Wave-Mode SAR vignettes over open sea, ~12 M images 2015–2024, VV, downsampled to 50 m	Yes — surface roughness only	256×256 px @ 50 m = 12.8 km; single acquisition; DINO / DINOv2 with dynamic dataset curation	ResNet50 2048-D • ViT-S/B; v2 ViT 384-D; v1 Apache 2.0 on HF	Sea-state: TenGeoP geophysical classes, wave height (RMSE 0.63–0.72 m), wind speed (RMSE $\sim 1.4 \text{ m s}^{-1}$), icebergs
Copernicus-FM / Copernicus-Pretrain (TUM, 2025)	S-1, S-2, S-3 OLCI (300 m), S-5P (1 km), DEM on a 0.25° ERA5 grid; 18.7 M images, ~312 k grid cells	Partial — 'whole land surface and near-land ocean'	per-modality patches inside 0.25° cells; dynamic hypernetwork accepts any spectral/non-spectral input + metadata	ViT encoders; code, data and weights released (github zhu-xlab); Copernicus-Bench, 15 tasks	Unified multi-sensor EO backbone; coastal seas incidental, not a target
Weather/climate FMs (Prithvi WxC, Aurora, ClimaX, AtmoRep)	Reanalysis grids (MERRA-2, ERA5) whose surface fields exist over ocean cells; Aurora has an ocean-wave fine-tune (HRES-WAM)	Ocean cells present, but as reanalysis background — SST/ocean state not the learning target	0.25° – 0.5° global grids; 1–2 lead-time steps	Open weights (varies); no ocean embedding product	Atmospheric forecasting; wave forecasting (Aurora)
Location encoders (SatCLIP, GeoCLIP)	Coordinates $\leftrightarrow$ imagery/photos; SatCLIP samples S-2 globally, GeoCLIP uses geotagged photos	Effectively no — sea coordinates unverified or absent	point location $\rightarrow$ vector	Open	Geographic priors, not ocean state

Verdict. Two purpose-built ocean encoders exist besides ours — both surface, both single-snapshot, one optical (colour + SST) and one radar (sea state). Everything else masks the sea out or carries it as reanalysis background; forecast-only systems (GLONET, Xihe, WenHai, AI-GOMS, Samudra) learn ocean dynamics but publish forecasts, not reusable embeddings. Earth 2 is, to our knowledge, the only embedding model whose inputs include the ocean interior through time.

Who embeds the ocean — and how deep?

This slide answers a question we kept returning to: which representation models have ever seen the sea in their training data, and how deep do they go?

Earth 2 (ours) is in the first row because it is the only entry whose inputs include the ocean interior — temperature and salinity at depth from Argo floats (the global fleet of autonomous profiling instruments), plus sea-surface height, currents and wind at the surface. Our stencil is a 3-by-3 patch of quarter-degree cells, one monthly or five-day snapshot, with 24 months of history in the second-stage forecasting model. Output: 64 numbers per pixel-month.

Granite-Geospatial-Ocean (previous slide) and OceanSAR are the two published encoders trained deliberately on sea pixels — but both are surface-only. Granite sees ocean colour and temperature; OceanSAR (from the company Galeio) sees Sentinel-1 radar "wave mode" images of the sea surface, about 12 million of them, and predicts sea state, wave height and wind speed.

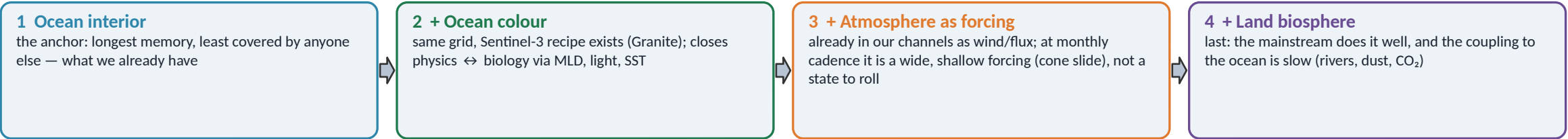
Copernicus-FM (TU Munich) covers "the whole land surface and near-land ocean" on a quarter-degree grid — coastal seas are incidental. Weather and climate models (Prithvi WxC, Aurora, ClimaX) contain ocean grid cells only because reanalysis fields are global; the sea is background, not the learning target. Location encoders such as SatCLIP work on coordinates, not ocean state.

Verdict: nothing else embeds the ocean interior through time. Forecast systems such as GLONET or Samudra learn ocean dynamics but publish forecasts, not reusable embeddings.

Four interacting spheres — who embeds which

Atmospheric weather · ocean weather (incl. the interior) · ocean biosphere · land biosphere. Nobody represents two of them jointly today.

Sphere	Who embeds it today	Native cadence · memory	Observing geometry	Couples to the others through
a) Atmospheric weather	Forecast FMs: Prithvi WxC, Aurora, ClimaX, GraphCast/AIFS/Pangu (states, not reusable embeddings). AlphaEarth ingests ERA5-Land only as an input.	hours–days · ~10 d predictability; slow modes (ENSO, NAO) months	dense gridded reanalysis; radiances and stations upstream	wind stress, heat & freshwater flux → ocean; precipitation, radiation → both biospheres
b) Ocean weather	Surface only: OceanSAR (sea state), Granite (SST as one band), AIFS-ocean 2026 (SST/SSS/SSH/currents, forecast). Interior: Earth 2 — the only embedding with T, S at depth.	days (surface) to decades (interior) · mixed layer months, interior years; top 2.5 m hold the heat of the whole atmospheric column	gridded SST/SSH from satellites; sparse Argo profiles below; nothing under ice	SST → atmosphere; mixed-layer depth, nutrients, light → ocean biosphere; carbon uptake → land via CO ₂ (slow)
c) Ocean biosphere	Granite-Geospatial-Ocean (colour → chl-a, primary production); carbon-sink mappers SOM-FFN, CSIR-ML6; CANYON-B for BGC-Argo — all task models, one FM	days–seasons · blooms weeks, seasonal cycle, decadal sink trends	cloud-gapped optical swaths (Sentinel-3, MODIS); BGC-Argo sparse; pCO ₂ ship tracks	phytoplankton ← MLD, SST, wind (physics); CO ₂ flux → atmosphere; export → interior carbon
d) Land biosphere / surface	The mainstream: AlphaEarth, TESSERA, OlmoEarth, TerraMind, Prithvi-EO 2.0 — all fold a year, none touch the sea	seasonal–annual · phenology, multi-year land-use memory	dense 10–30 m tiles, frequent revisits; LiDAR, SAR	runoff, dust, nutrients → coastal ocean; evapotranspiration, albedo → atmosphere



Three design consequences. (i) Memories differ by orders of magnitude, so each sphere needs its own (v, τ) cone and the coupling variables (SST, fluxes, chlorophyll–light–MLD) are the L-shaped unions. (ii) Biospheres run on the calendar, physics on the clock: a joint model needs period tokens (pixel-year, slide on the TESSERA arm) for c) and d) and dated-state tokens for a) and b). (iii) The observing systems differ in geometry — grids, sparse profiles, cloud-gapped swaths, 10 m tiles — and our masked-vs-never-measured token distinction is the primitive that lets one encoder take all of them.

Four interacting spheres — who embeds which

Yannick's framing of the long-term goal: the Earth system has four interacting "spheres", and eventually we want one representation that spans them all. This slide maps who covers which today and proposes an order for joining them.

a) Atmospheric weather: covered by forecast models (Prithvi WxC, Aurora, GraphCast and others) that produce states, not reusable embeddings. Native timescale hours to days; the atmosphere is predictable for about ten days.

b) Ocean weather: the surface is covered by OceanSAR, Granite and ECMWF's 2026 AIFS ocean extension; the interior only by Earth 2. Timescales run from days at the surface to decades in the deep. A physical fact drives everything: the top two and a half metres of ocean hold as much heat as the entire atmosphere above it, so the ocean is the slow, remembering partner.

c) Ocean biosphere: phytoplankton, ocean colour, carbon uptake. Granite plus task-specific carbon-flux models (SOM-FFN, CSIR-ML6, CANYON-B). Timescales: blooms in weeks, seasons, decadal trends.

d) Land biosphere: the entire mainstream of Earth observation models, all of which fold a year and none of which touch the sea.

Nobody represents two spheres jointly. The proposed sequence: start from the ocean interior (longest memory, least covered by anyone else — what we already have), add ocean colour (same grid, Granite's recipe exists, and it closes the physics-to-biology loop through mixed-layer depth, light and temperature), then treat the atmosphere as a forcing (already in our channels as wind and fluxes), and add land last.

Three design consequences follow: each sphere needs its own speed-and-memory "cone" (introduced on slide 26); biospheres run on the calendar so they want year-summary tokens while physics wants dated snapshots; and the observing systems differ in shape (grids, sparse profiles, cloud-gapped swaths, 10-m tiles), which our distinction between "hidden from the model" and "never measured" tokens is built to handle.

Head-to-head I — frozen embeddings, one protocol

OlmoEarth paper, Table 2: kNN (k = 20, cosine) for single-image classification, linear probe for segmentation and multi-temporal tasks. Five of our six appear.

Model (frozen encoder)	BigEarthNet μ F1	So2Sat acc	EuroSAT acc	CropHarvest-Togo S1+S2 acc	m-cashew mIoU	SA-crop mIoU	PASTIS (S2) mIoU	MADOS mIoU	Sen1Floods11 mIoU
OlmoEarth Base (89 M)	62.4	67.7	94.7	82.0	32.3	28.9	50.6	67.2	79.2
OlmoEarth Large (308 M)	62.0	68.2	96.3	79.7	30.9	28.5	51.8	66.4	79.8
TerraMind Base	63.9	46.7	85.6	77.5	46.0	30.4	40.9	66.0	78.7
TerraMind Large	63.9	47.4	90.0	75.2	50.4	31.2	41.3	67.5	78.4
Prithvi-EO 2.0 300M	51.6	34.7	82.2	–	46.8	24.7	37.2	50.9	–
Prithvi-EO 2.0 600M	51.0	34.0	81.2	–	45.8	26.6	37.5	52.2	–
TESSERA (precomputed)	–	–	–	81.0	–	–	–	–	–
Galileo Base	58.3	55.7	92.8	77.5	28.9	25.3	39.6	68.4	79.4
DINOv3-Sat 7B	61.6	50.1	91.3	–	54.1	31.7	26.3	59.7	–
Panopticon Base	64.9	60.5	95.2	76.5	32.7	27.3	30.2	66.1	78.0
Copernicus-FM Base	64.6	50.3	84.7	70.6	32.2	28.4	32.1	63.9	77.6
CROMA Large	59.2	48.2	85.8	81.0	27.0	30.4	42.7	66.4	78.8

Reading the table. Green = best in column (all rows). Shaded rows are the models in this deck; grey rows are context. On frozen features OlmoEarth leads the classification columns (So2Sat, EuroSAT, Sen1Floods11, PASTIS), TerraMind leads BigEarthNet and the cashew/SA-crop segmentation, and Prithvi-EO 2.0 trails on every classification column but is competitive on m-cashew. TESSERA appears once (CropHarvest-Togo, 81.0 — within 1 pp of OlmoEarth Base). AlphaEarth is not in this table (its embeddings are annual and 10 m, so the authors compared it separately — next slides). Caveat: authored by the OlmoEarth team.

Head-to-head I — frozen embeddings, one protocol

The first of four slides with actual benchmark numbers, transcribed from the papers. This one uses the widest comparison available: Table 2 of the OlmoEarth paper, which evaluates many models with a single frozen-embedding protocol.

"Frozen" means the encoder is not retrained; only a trivial classifier sits on top. kNN (k-nearest neighbours) assigns a test image the majority label of its 20 nearest training embeddings — the purest test of embedding quality. "Linear probe" fits a single linear layer, used for segmentation and multi-temporal tasks.

Columns are benchmark datasets: BigEarthNet and So2Sat (land-cover classification), EuroSAT (scene classification), CropHarvest-Togo (crop vs non-crop), m-cashew and SA-crop (crop segmentation), PASTIS (crop parcels through time), MADOS (marine debris and oil), Sen1Floods11 (flood mapping from radar). Scores are accuracy or micro-F1 for classification and mIoU for segmentation — all 0 to 100, higher is better.

Shaded rows are the models in this deck; grey rows are other models for context. Green is the best in each column across all rows.

What it says: on frozen features OlmoEarth leads the classification columns by large margins (So2Sat 68 vs 47 for TerraMind vs 35 for Prithvi), TerraMind leads BigEarthNet and the cashew and SA-crop segmentation, and Prithvi trails on every classification column. TESSERA appears once, on CropHarvest-Togo, within one point of OlmoEarth. AlphaEarth is absent from this table because its embeddings are annual and could not be run under this protocol.

Caveat, which applies to every table in this section: it was authored by one of the contestants.

Head-to-head II — fine-tuning and PANGAEA

Top: OlmoEarth Table 3 (full fine-tuning, encoder unfrozen after 20 % of epochs). Bottom: TerraMind Table 6 (PANGAEA, frozen encoder + UPerNet, mIoU).

Fine-tuned — accuracy / mIoU (OlmoEarth paper, Table 3)

Model (fine-tuned)	BigEarthNet	So2Sat	EuroSAT	m-cashew	SA-crop	PASTIS	MADOS	Sen1Floods11
OlmoEarth Base	72.0	68.6	98.7	79.8	39.6	64.3	77.8	79.8
OlmoEarth Large	72.4	68.1	98.5	80.6	40.8	66.3	81.8	79.8
TerraMind Base	72.6	66.1	97.6	80.9	39.2	59.9	73.2	79.5
TerraMind Large	74.0	65.4	97.8	81.3	41.1	60.9	71.5	79.5
Prithvi-EO 2.0 600M	70.6	64.7	96.8	81.1	38.8	58.6	69.3	-
Galileo Base	69.2	64.7	97.8	78.8	35.7	61.2	71.9	79.7
Copernicus-FM	71.3	66.8	98.5	78.7	33.6	54.6	66.0	78.6
OlmoEarth Base, random init (no pretraining)	61.0	48.9	80.3	43.0	27.5	43.9	45.6	77.0

PANGAEA — mIoU, frozen encoder + UPerNet (TerraMind paper, Table 6; Prithvi appears only as v1.0)

Model	HLS Burn Scars	MADOS	PASTIS-R	Sen1Floods11	FiveBillionPx	DynamicEarthNet	CropType S.Sudan	SpaceNet 7	AI4SmallFarms	Average	Avg rank
TerraMind Large	82.93	75.57	43.13	90.78	63.38	37.89	55.04	59.98	27.47	59.57	3.44
TerraMind Base	82.42	69.52	40.51	90.62	59.72	37.87	55.80	60.61	28.12	58.35	3.94
Prithvi-EO 1.0 100M	83.62	49.98	33.93	90.37	46.81	27.86	43.07	56.54	26.86	51.00	11.00
CROMA	82.42	67.55	32.32	90.89	51.83	38.29	49.38	59.28	25.65	55.29	6.61
DOFA	80.63	59.58	30.02	89.37	43.18	39.29	51.33	61.84	27.07	53.59	8.22
U-Net trained from scratch	84.51	54.79	31.60	91.42	60.47	39.46	47.57	62.09	46.34	57.58	4.89

Reading. Once fine-tuned, the gap closes: TerraMind Large and OlmoEarth Large split the columns (TerraMind: BigEarthNet, cashew, SA-crop; OlmoEarth: So2Sat, EuroSAT, PASTIS, MADOS) and Prithvi-EO 2.0 600M sits 1–3 pp behind on the classification columns but 8–13 pp behind on PASTIS and MADOS. The random-init row is the size of the pretraining effect itself (+10 to +37 pp). On PANGAEA, TerraMind is the only foundation model that beats a U-Net trained from scratch on average (59.6 vs 57.6); Prithvi-EO 1.0 loses to it. Prithvi-EO 2.0’s own GEO-Bench table (600M-TL best on 6 of 12 sets vs DOFA / Scale-MAE / DeCUR / Satlas) contains none of the other models here.

Head-to-head II — fine-tuning and PANGAEA

Two more tables. The top one is OlmoEarth's Table 3: the same models, now fully fine-tuned — the whole network is retrained for each task (the encoder is unfrozen after the first 20 % of training). The bottom one is TerraMind's Table 6 on the PANGAEA benchmark, a community suite of about eleven segmentation, change-detection and regression tasks, evaluated with a frozen encoder plus a standard segmentation decoder (UPerNet).

Top table: once fine-tuned, the gap between models closes. TerraMind Large and OlmoEarth Large split the columns roughly evenly, and Prithvi-EO 2.0 (600 M) sits one to three points behind on the classification columns but eight to thirteen points behind on the PASTIS and MADOS segmentation columns. The last row is the same OlmoEarth architecture trained from random initialisation — no pretraining at all. The distance between that row and the others (10 to 37 points) is the value of pretraining itself.

Bottom table: mIoU per dataset plus an average and an average rank. TerraMind Large is the only foundation model whose average (59.6) beats a U-Net trained from scratch (57.6). Prithvi appears only as version 1.0 and comes last. HLS Burn Scars, FiveBillionPixels, DynamicEarthNet, SpaceNet 7 and AI4SmallFarms are the other PANGAEA datasets.

Not shown: Prithvi-EO 2.0's own GEO-Bench table, because none of the other models in this deck appear in it.

Head-to-head III — the two embedding products

TESSERA paper, Table 1 (frozen embeddings + light head, 100 % labels unless noted) · OlmoEarth paper, Table 7 · an independent Local-Climate-Zone study (Bern)

TESSERA vs AlphaEarth vs frozen RSFMs (TESSERA paper) — F1 / mIoU ↑ , RMSE ↓

Model	TreeSatAI-TS F1	Austrian crop F1 (30 % labels)	PASTIS-R mIoU	Austrian crop seg. mIoU	BioMassters RMSE (t/ha)	Borneo canopy RMSE (m)
TESSERA (128-D, frozen)	77.96	82.09	50.68	53.12	27.43	12.21
AlphaEarth (64-D, frozen)	76.90	56.36	51.08	25.70	29.59	16.11
Prithvi-EO (frozen)	65.86	–	34.09	16.25	41.12	19.71
Presto (frozen)	67.81	57.89	–	34.04	–	17.88
Galileo (frozen)	69.44	–	27.92	22.80	38.52	20.48
SkySense (frozen)	69.35	–	32.87	26.82	32.52	16.56
U-Net baseline (frozen features)	71.39	–	30.16	27.00	36.41	17.79

Reading. TESSERA beats AlphaEarth on 5 of these 6 tasks in TESSERA's own table (AlphaEarth edges PASTIS-R by 0.4 mIoU), with the largest margins on pixel-wise crop tasks where per-pixel time series matter most; both precomputed products beat every frozen ViT-style RSFM including Prithvi by wide margins. Against OlmoEarth, AlphaEarth wins only where OlmoEarth is also frozen (Nandi frozen+decoder); once OlmoEarth is fine-tuned it wins all five — and AlphaEarth cannot be fine-tuned. The independent Bern study puts the two products within 1 pp of each other and 2–5 pp above raw composites. AlphaEarth's own paper reports a 23.9 % average error reduction vs the next-best method over 15 evaluations (per-dataset values are charts only), losing on just one (LCMAP land-use change).

AlphaEarth vs OlmoEarth Base (OlmoEarth paper, Table 7)

Model · protocol	Nandi acc	AWF acc	Ecosystem acc	LFMC L1 ↓	Solar farm mIoU
AlphaEarth · kNN	55.6	81.0	60.6	–	–
AlphaEarth · frozen + decoder	66.0	75.9	61.2	23.1	77.5
AlphaEarth · full fine-tune	n/a	n/a	n/a	n/a	n/a
OlmoEarth · kNN	66.2	82.0	59.3	–	–
OlmoEarth · frozen + decoder	62.9	84.0	61.1	19.9	84.8
OlmoEarth · full fine-tune	82.2	86.0	62.4	17.9	86.7

Independent: LCZ mapping, Bern (arXiv 2606.20034)

Input to Attention U-Net	Accuracy	IoU
Sentinel-1 + 2 composites	0.87	0.77
AlphaEarth 64-D	0.89	0.81
TESSERA 128-D	0.90	0.82

Head-to-head III — the two embedding products

This slide compares the two "products" — the models whose main deliverable is a precomputed global embedding layer: AlphaEarth and TESSERA.

Left table, from the TESSERA paper: frozen embeddings plus a light head, on six tasks. F1 and mIoU are higher-is-better (arrows up); RMSE — root-mean-square error, in tonnes per hectare for biomass and metres for canopy height — is lower-is-better (arrows down). TESSERA beats AlphaEarth on five of six, with the largest margins on pixel-wise crop tasks where a per-pixel time series matters most (Austrian crop F1 82 vs 56). AlphaEarth edges PASTIS-R by 0.4 points. Both products beat every frozen ViT-style model by wide margins.

Top-right table, from the OlmoEarth paper: AlphaEarth versus OlmoEarth under three protocols. With both frozen, results are mixed. Fine-tuned OlmoEarth wins all five tasks — and the "n/a" row is the point: AlphaEarth cannot be fine-tuned, because only the embeddings, not the network, are released. LFMC is live fuel moisture content, an L1 (absolute) error, lower is better.

Bottom-right, an independent study: mapping Local Climate Zones in Bern with an Attention U-Net fed either raw Sentinel composites, AlphaEarth, or TESSERA. The two products are within one point of each other and two to five points above raw imagery.

AlphaEarth's own paper reports a 23.9 % average error reduction versus the next-best method across 15 evaluations, losing only on land-use change; its per-task values are published as charts without numbers, which is why they are not tabulated here.

Head-to-head IV — the ocean encoders

Granite-Geospatial-Ocean, Table 1 (5-fold CV RMSE, log₁₀ units) · OceanSAR-1, Tables 2–3 (Sentinel-1 wave-mode tasks)

Granite-Geospatial-Ocean — fine-tuned FM vs from scratch vs random forest (RMSE ↓)

Model	Chl-a RMSE log ₁₀ mg m ⁻³	Primary production RMSE log ₁₀ mgC m ⁻² d ⁻¹
OLCI + SST · foundation model, fine-tuned	0.14	0.39
OLCI only · foundation model, fine-tuned	0.16	0.39
OLCI + SST · same net from scratch	0.16	0.42
OLCI only · same net from scratch	0.16	0.43
Random forest on pixel values	0.16	0.40

± spreads: chl-a 0.03–0.10, PP 0.04–0.07 across folds — the FM gain (0.02 / 0.03–0.04) is inside the fold spread; the authors' stronger claim is label-efficiency (gain grows as labels shrink to 25 %) and spatial pattern (SSIM vs the operational L2 product: FM 0.88, scratch 0.82, decision tree 0.68). No comparison to OC4Me-type algorithms is given.

OceanSAR-1 — kNN on frozen features vs other SAR FMs (acc ↑, RMSE ↓)

Model	TenGeoP acc %	Wave height RMSE (m)	Wind speed RMSE (m s ⁻¹)
OceanSAR-1 ViT-B/8	83.6	0.63	1.37
OceanSAR-1 ViT-S/8	82.1	0.64	1.38
OceanSAR-1 ResNet50	75.5	0.75	1.62
SoftCon ViT-B/14	74.8	1.01	2.08
CROMA ViT-B/8	65.4	0.78	1.95
DOFA ViT-L/16	63.4	0.95	2.43
MoCo ResNet50	60.9	0.83	1.98

With linear-probe / LoRA fine-tuning OceanSAR-1 ViT-B/8 reaches 88.3 % / 0.54 m / 1.29 m s⁻¹ (MoCo-FT 86.5 / 0.77 / 1.80). Ocean-only pretraining beats land-trained SAR FMs by 9–23 pp on sea-state classification and cuts wind-speed error by 30–45 %.

Why no cross-model ocean table. The two ocean encoders measure different physics (optical colour vs radar roughness) against different in-situ truths, and neither has been run on a land benchmark or on each other's tasks. The only common denominator with the land models is the protocol — frozen features + small head, and 'FM vs same net from scratch' as the pretraining control — which is also the protocol of our attribution matrix. Our numbers (RAPID r 0.617 pixel / 0.672 patch vs raw 0.613 / 0.659) are the same kind of statement as Granite's 0.14 vs 0.16: a pretraining margin measured against a matched from-scratch control, and in both cases a small one.

Head-to-head IV — the ocean encoders

The two ocean encoders cannot be compared with the land models — no shared benchmark exists — so this slide shows each against its own baselines.

Left: Granite-Geospatial-Ocean, Table 1. Five-fold cross-validated RMSE in log10 units for chlorophyll-a and primary production, comparing the fine-tuned foundation model with the same network trained from scratch and with a random forest on raw pixel values. The foundation model is best in both columns, but the margins (0.02 and 0.03) are smaller than the spread between folds. The authors' stronger claims are label efficiency (the advantage grows as training labels shrink) and better spatial pattern (SSIM — structural similarity — of 0.88 versus 0.82 against the operational product).

Right: OceanSAR-1, kNN on frozen features against other SAR foundation models. TenGeoP is a ten-class geophysical-phenomena dataset (accuracy, higher is better); wave height and wind speed are RMSE in metres and metres per second (lower is better). Ocean-only pretraining beats land-trained SAR models by 9 to 23 points on classification and cuts wind-speed error by 30 to 45 %.

Bottom box: the only thing the ocean and land models share is the protocol — frozen features plus a small head, and "foundation model versus same network from scratch" as the control for the value of pretraining. That is also exactly what our attribution matrix does: our pixel and patch codecs versus raw data at matched receptive field. Our pretraining margin (0.617 vs 0.613 correlation for a single pixel; 0.672 vs 0.659 for the 3-by-3 patch) is the same kind of statement as Granite's 0.14 versus 0.16 — a small one.

Who wins where — a digest of the tables

Rankings restricted to the models in this deck, per protocol; every source table was written by one of the contestants

Protocol · metric family	Ranking among our models	Margin / evidence	Source & caveat
Frozen features · classification accuracy / μ F1	OlmoEarth > TerraMind > Prithvi-EO 2.0	So2Sat 68 vs 47 vs 35; EuroSAT 96 vs 90 vs 82; BigEarthNet TerraMind 63.9 edges OlmoEarth 62.4	OlmoEarth Table 2 (own paper)
Frozen features · segmentation mIoU	split: TerraMind (cashew 50.4, SA-crop 31.2) · OlmoEarth (PASTIS 51.8, MADOS 67.2, floods 79.8) · Prithvi last	Prithvi-EO 2.0 competitive only on m-cashew (46.8)	OlmoEarth Table 2
Fine-tuned · accuracy / mIoU	TerraMind-L $\approx$ OlmoEarth-L > Prithvi-EO 2.0 600M	columns split 3/4 between the two leaders; Prithvi 1–3 pp behind on classification, 8–13 pp on PASTIS/MADOS; pretraining itself worth +10 to +37 pp (random-init row)	OlmoEarth Table 3
PANGAEA · mIoU, frozen + UPerNet	TerraMind-L 59.6 > TerraMind-B 58.4 > U-Net scratch 57.6 > ... > Prithvi-EO 1.0 51.0	only GFM above the from-scratch U-Net; Prithvi 2.0 not evaluated	TerraMind Table 6
GEO-Bench-2 · capability ranks (of 14)	Prithvi 600M-TL core rank 5 · TerraMind-L 6 · Prithvi 300M-TL 9 · TerraMind-B 11; TerraMind-L rank 1 on multi-temporal and multi-spectral	ranks only, no raw scores	GEO-Bench-2, arXiv 2511.15658 (independent); no AlphaEarth / TESSERA / OlmoEarth / Granite
Precomputed products · frozen + head	TESSERA > AlphaEarth on 5/6 TESSERA tasks; $\approx$ tie on Bern LCZ (0.82 vs 0.81 IoU); both $\gg$ frozen Prithvi	Austrian crop F1 82 vs 56; canopy RMSE 12.2 vs 16.1 m; PASTIS AlphaEarth +0.4	TESSERA Table 1 (own) · LCZ study (independent)
Product vs fine-tunable model	fine-tuned OlmoEarth > AlphaEarth on 5/5; frozen-vs-frozen mixed (AlphaEarth wins Nandi 66.0 vs 62.9)	Nandi 82.2 vs 66.0; solar farm 86.7 vs 77.5	OlmoEarth Table 7 (own)
Ocean · RMSE vs in-situ, FM vs scratch	Granite: 0.14 vs 0.16 (chl-a), 0.39 vs 0.42 (PP) · OceanSAR-1: 83.6 % vs 74.8 % best land-trained SAR FM	Granite gain inside fold spread; OceanSAR gain large	own papers; no shared task with land models

Two structural caveats. No benchmark contains all six models — the widest single table (OlmoEarth's) has five and omits AlphaEarth; the only independent multi-model source (GEO-Bench-2) has two. And no public benchmark has an ocean-state, transport or interior task, so none of these rankings transfer to our problem; the transferable lesson is the protocol — frozen probe vs fine-tune, and a from-scratch control at matched receptive field, which is what our attribution matrix already does.

Who wins where — a digest of the tables

A one-page digest of the four head-to-head slides: for each evaluation protocol and metric family, how the models in this deck rank, with the margin and the source.

Reading down: with frozen features OlmoEarth leads classification and shares segmentation with TerraMind; fine-tuned, TerraMind Large and OlmoEarth Large tie with Prithvi behind; on PANGAEA TerraMind is the only model above a from-scratch U-Net; GEO-Bench-2 (an independent 2025 benchmark that publishes ranks, not scores) places Prithvi 600M-TL fifth and TerraMind Large sixth of fourteen; among the two products TESSERA beats AlphaEarth on five of six tasks in TESSERA's own table but they tie in an independent study; a fine-tuned OlmoEarth beats AlphaEarth everywhere because AlphaEarth cannot be fine-tuned; and the ocean models are measured only against themselves.

The two structural caveats at the bottom are the honest summary. First, no benchmark contains all six models — the widest single table has five and omits AlphaEarth, and the only independent multi-model source has two. Second, no public benchmark has an ocean-state, transport or interior task, so none of these rankings transfer to our problem. What transfers is the protocol: frozen probe versus fine-tune, and a from-scratch control at matched receptive field, which our attribution matrix already implements.

What this means for our ocean representation work

Six models, two design axes, one gap

1 Axis 1 — pixel vs. patch

TESSERA proves a per-pixel encoder with zero spatial context is competitive on land; every other model buys context with a ViT patch grid. We have already priced this axis: the attribution matrix reads single-pixel raw 0.613 vs codec 0.617, and 3×3 raw 0.659 vs codec 0.672 (two seeds) — the neighbourhood is worth $\approx +0.05$, pretraining $\approx +0.01$.

2 Axis 2 — snapshot vs. period

TerraMind and Granite are snapshots; Prithvi and OlmoEarth stack dated frames; AlphaEarth and TESSERA fold a full year into one vector. Our stage-1 codec is a snapshot of a monthly (or 5-day) mean and the month-block codec folds only ~ 6 pentads. The year-folded design is the untested axis — hence the proposed arm on the last slide.

3 Product vs. model

AlphaEarth is a dataset you cannot re-run; TESSERA and OlmoEarth ship both weights and embeddings. For the ocean nothing precomputed exists — whatever we release is the first embedding-field product for the ocean interior.

4 The gap

Only two published encoders are trained on sea pixels — Granite-Geospatial-Ocean (colour + SST) and OceanSAR (sea-state radar) — and both are surface, single-instant models. To our knowledge no published model embeds the gridded ocean state (T, S, SSH, currents) through time — the territory our codec occupies.

What this means for our ocean representation work

Four takeaways for Earth 2, each tied to something measured.

Axis 1, pixel versus patch. TESSERA shows a per-pixel encoder with no spatial context is competitive on land; every other model buys context with a patch grid. We have already priced this axis in our attribution matrix: reading the RAPID transport with a single-pixel codec gives a correlation of 0.617 (raw data 0.613); with the 3-by-3 patch, 0.672 averaged over two seeds (raw 0.659). So the neighbourhood is worth about +0.05, and pretraining about +0.01. (RAPID is the mooring array at 26.5°N that measures the Atlantic overturning circulation; "correlation" is how well our read-out tracks it.)

Axis 2, snapshot versus period. TerraMind and Granite are snapshots; Prithvi and OlmoEarth stack dated frames; AlphaEarth and TESSERA fold a full year into one vector. Our first-stage codec is a snapshot of a monthly (or five-day) mean, and the month-block codec folds only about six five-day bins. The year-folded design is the axis we have not tested — hence the proposed experiment on the next slide.

Product versus model. AlphaEarth is a dataset you cannot re-run; TESSERA and OlmoEarth ship both. For the ocean nothing precomputed exists at all.

The gap. Only two published encoders are trained on sea pixels — Granite and OceanSAR — and both are surface, single-instant models. To our knowledge no published model embeds the gridded ocean state through time. That is the territory our codec occupies.

Proposed arm: a TESSERA-style pixel-year codec

Fold the year, not the neighbourhood — the one TESSERA ingredient the programme has not yet tested

Why not a 1×1 spatial ablation Already done. The attribution matrix has $p = 1$ and 3×3 at both raw and codec: 0.613 / 0.617 vs 0.659 / 0.672. The spatial axis is priced; the temporal axis is not.

What TESSERA does that we don't One embedding per pixel-YEAR: a 4-layer temporal transformer + GRU pooling over ~40 dated observations of a single pixel, day-of-year embeddings, Barlow Twins objective (no reconstruction), int8 QAT. Our stage-1 z is one pixel-MONTH; stage 2 then runs over the monthly z 's.

Arm A — pixel-year codec Encoder over the 12 (or 24) pixel-months of one $\frac{1}{4}^\circ$ pixel: channel tokens carry month-of-year; $p = 1$ to isolate the temporal fold; emit one $z \in \mathbb{R}^{64}$ per pixel-year (or rolling 12-month window ending at t). Decoder queries (channel, Δmonth) up to ± 12 . Token count $\approx 12 \times (C+2) \approx 310$ at 24 channels — same order as the 282-token month-block codec, so it fits the existing recipe.

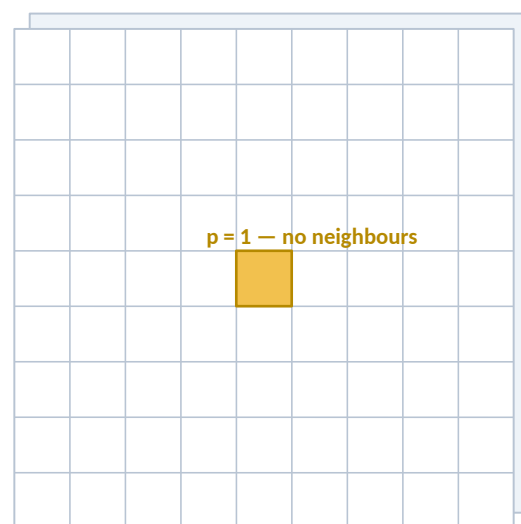
Arm B — objective swap Same encoder, Barlow Twins on two random date-subsamples of the same pixel-year (TESSERA's augmentation) instead of masked reconstruction. Tests whether a non-reconstructive objective keeps the deep-density detail the decoder had to be re-taught to read.

Controls (matched) Mean of the 12 monthly z 's (pooling control) · stage-2 over monthly z 's (current path) · raw 12-month stack into the same attention head (end-to-end control, as in the attribution matrix). Two codec seeds minimum — head numbers are never quoted from one.

Read-outs Probe ladder → RAPID r and RMSE (Sv); round-trip variance lost per channel; rolled corridor AUC at $h = 12$ with a year-folded z as the state. Success = the folded z beats the pooled control by more than the two-seed spread.

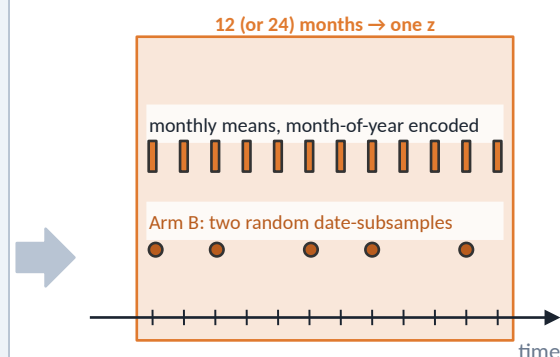
Stencil — what feeds one embedding

SPACE



Output pixel: one $\frac{1}{4}^\circ$ cell (≈ 28 km).
Receptive field = the same cell; the 3×3 patch is deliberately switched off so only the time axis changes.
Sheets: the 24–25 ocean channels (T, S, SSH, wind, ...).

TIME



A period embedding of the ocean state, TESSERA-style, instead of a snapshot of one monthly mean.
Moves our point on the space × time map from 'single snapshot' to '1 year folded'.

OUTPUT



Proposed arm: a TESSERA-style pixel-year codec

A concrete experiment proposal, framed honestly. An earlier draft of this deck suggested a "1-by-1 pixel-only control"; it turned out our paper already contains it (the attribution matrix has both single-pixel and 3-by-3 versions of raw and codec inputs). So the spatial question is settled; what TESSERA adds that we have not tried is the temporal fold.

What TESSERA does that we don't: it produces one embedding per pixel-YEAR from about 40 dated observations of one pixel, using a temporal transformer and day-of-year encodings, trained with Barlow Twins (no reconstruction) and quantised to int8 during training. Our first-stage vector z describes one pixel-MONTH; a second-stage model then runs over the sequence of monthly vectors.

Arm A — the pixel-year codec: an encoder that reads the 12 (or 24) pixel-months of one quarter-degree cell at once, with month-of-year tokens, and emits one 64-number vector per pixel-year. We keep the patch size at 1 so that only the time axis changes. Token count about 12 times (channels + 2), roughly 310 at 24 channels — the same order as our existing 282-token month-block codec, so it fits the current training recipe.

Arm B — swap the training objective: same encoder, but Barlow Twins on two random subsets of dates from the same pixel-year, instead of masked reconstruction.

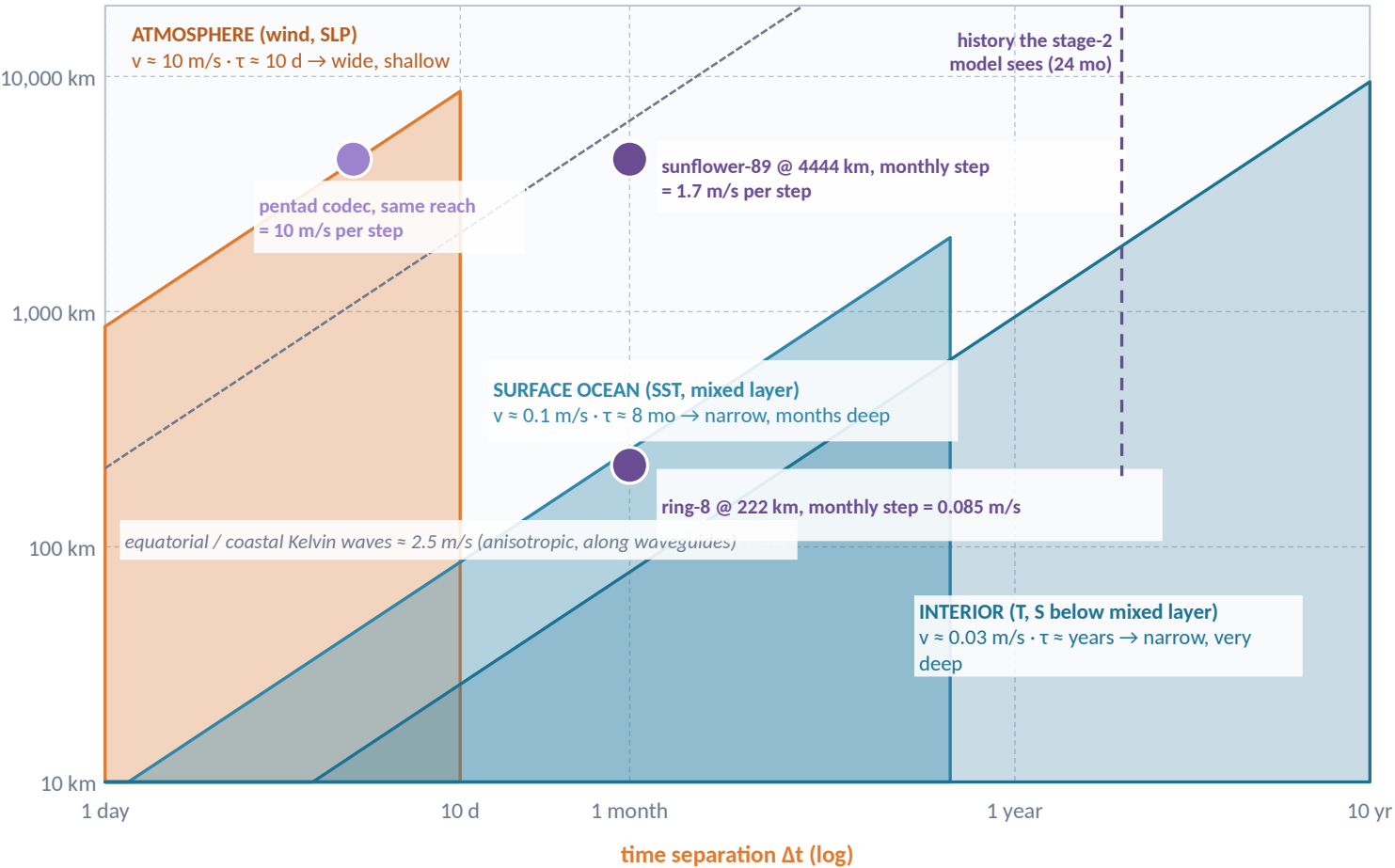
Controls: the average of the 12 monthly vectors (does folding beat pooling?), the current stage-2 pathway, and the raw 12-month stack fed to the same attention head (the end-to-end control from the attribution matrix). At least two codec seeds, because head numbers from one seed have proven noisy.

Read-outs: our probe ladder to the RAPID transport (correlation and RMSE in Sverdrups), the fraction of variance lost on the round trip, and the 12-month rolled corridor AUC — our standard forecast skill score. Success means the folded vector beats the pooled control by more than the seed spread.

A velocity-specific dependency cone per channel

What can influence one pixel Δt ahead is bounded by how fast each process propagates (v) and how long it remembers (τ): $\Delta x \leq v \cdot \Delta t$, $\Delta t \leq \tau$

distance Δx (log)



The model

For a process p with signal speed v_p and memory τ_p , the state at $(x, t+\Delta t)$ can depend on the past only inside
 $C_p(\Delta t) = \{ (\Delta x, \ell) : \Delta x \leq v_p (\Delta t + \ell), \Delta t + \ell \leq \tau_p \}$
 Δt is the forecast step (we predict $t+\Delta t$ from data up to now, t); ℓ is the lag of an input — it was observed at $t-\ell$, so it sits $\Delta t+\ell$ before the target ($\ell = 0$ is the newest input). $\Delta x \leq v \cdot (\Delta t + \ell)$ is the domain of dependence (the CFL / light-cone argument); $\Delta t + \ell \leq \tau$ says that beyond the process's own predictability its history is noise, not signal.

The union rule

A channel c is forced by several processes, so its cone is the union: $C_c = \bigcup_{p \in P(c)} C_p$. Temperature is forced by the atmosphere (fluxes) and by currents (advection), so $C_{\text{SST}} = C_{\text{atm}} \cup C_{\text{ocean}}$ — wide-and-shallow plus narrow-and-deep, which is not a cone but an L-shape in $(\Delta x, \Delta t)$.

Where our stencils sit

A stencil is a cylinder — the same reach at every lag. ring-8 @ 222 km per month is an ocean-advection speed (0.085 m/s); sunflower-89 @ 4444 km is 1.7 m/s, the speed of the fast ocean waveguides and inside the atmosphere's cone at $\Delta t = 1$ month. The rolled forecast rebuilds the cone recursively (dependency-cone note, §1), so reach-per-step $\times$ horizon is what the target finally sees.

Notation t = now (last observed step) · Δt = forecast step · ℓ = lag of an input (observed at $t-\ell$; $\ell = 0$ newest) · both in months here.

A velocity-specific dependency cone per channel

This slide introduces the physical idea Yannick proposed: what can influence a pixel Δt ahead is bounded by how fast each process propagates and how long it remembers.

Notation first. t is "now", the last observed step. Δt is the forecast step — we predict the target at $t + \Delta t$. ℓ (a script L) is the lag of an input: it was observed at $t - \ell$, so it sits $\Delta t + \ell$ before the target; $\ell = 0$ is the newest input. v is a process's signal speed; τ (tau) is its memory — how long it stays predictable.

The model: for a process p , an input at lag ℓ can matter only if it is close enough to have travelled to the target in time, $|\Delta x| \leq v \cdot (\Delta t + \ell)$, and only if the process still remembers, $\Delta t + \ell \leq \tau$. The first condition is the domain-of-dependence or "light-cone" argument from numerical physics (the CFL condition); the second says that beyond a process's predictability its history is noise.

The chart is log-log: time separation on the horizontal axis (1 day to 10 years), distance on the vertical (10 km to 20,000 km), so a constant speed is a straight diagonal. Each process is a region: the atmosphere (amber) is wide but shallow — 10 m/s but only 10 days of memory; the surface ocean (blue) is narrow but months deep — 0.1 m/s, 8 months; the interior (teal) is narrower still but years deep — 0.03 m/s. The dashed line is the Kelvin-wave speed along coasts and the equator, an exception that is fast but anisotropic.

The union rule: a channel forced by several processes has the union of their cones. Sea-surface temperature is forced by the atmosphere (heat flux) and by currents (advection), so its region is wide-and-shallow plus narrow-and-deep — an L-shape, not a cone.

Where our stencils sit: a fixed stencil is a cylinder — same reach at every lag. Our ring-8 at 222 km per month is an ocean-advection speed (0.085 m/s); our sunflower-89 at 4444 km is 1.7 m/s, inside the atmosphere's cone at a one-month step. The dashed vertical line marks the 24 months of history our second-stage model sees.

What the cone predicts — and how to test it on our stencil

Per channel group: dominant processes, the useful reach per monthly step, the useful history, and the falsifiable consequence

Channel group	Processes (v , τ)	Useful reach per monthly step	Useful history	Design consequence
10 m wind, SLP, fluxes	synoptic advection 10 m/s, $\tau \approx 5\text{--}10$ d; planetary modes (NAO, ENSO teleconnections) $\tau \approx$ months, forced by SST	causal reach in one month = the globe, but $\tau < \Delta t$: monthly wind is not predictable from its own history — only via slow ocean modes	1 step (its own); the ocean's history through SST	treat as wide, shallow forcing: large stencil, no depth. At pentad $\Delta t \approx \tau$ the wind becomes partly self-predictable — a pentad-only gain to look for
SST, mixed-layer T	surface currents 0.1–0.3 m/s (260–800 km/mo); air–sea forcing (atmospheric cone); mixed-layer memory $\tau \approx 3\text{--}12$ mo	advective 300–800 km, plus the atmospheric cone via forcing $\approx$ synoptic correlation length 1,000–3,000 km	several months (re-emergence up to a year)	union: needs BOTH the wide wind stencil and months of history — the one group that justifies the full cylinder
Interior T, S (below ML)	gyre advection 0.02–0.1 m/s (50–260 km/mo); midlatitude Rossby 0.02–0.05 m/s; $\tau \approx$ years–decades	50–300 km — a 3×3 patch at $\frac{1}{4}^\circ$ already covers it; 4444 km is 15–90 $\times$ too wide for one step	24 months and more; skill should keep rising with history	narrow, deep: small reach, long history. Reach can be cut with no loss — the concrete answer to the open question in the cone note (§5)
SSH	baroclinic Rossby 0.03 m/s (mid-lat) $\rightarrow \sim 0.3$ m/s (10°); Kelvin / coastal waves 1–3 m/s along waveguides; barotropic adjustment in hours (averaged out at monthly Δt)	interior: hundreds of km; along coasts and the equator: thousands of km in one step	months	anisotropic: an along-boundary / along-equator arm on the stencil, narrow elsewhere — a shape the sunflower approximates by brute force
Transport target (RAPID 26°N)	thermal wind = east–west density difference across the basin; set by interior Rossby adjustment (yrs) and by fast boundary waves (weeks)	the whole section, both boundaries	years for the interior arm; the current step for the boundary arm	why the pooled ridge failed and the unpooled attention head worked (paper Sladder): the target's cone is the section, not its mean

Four falsifiable predictions (each an ablation we can run)

- 1 Group-specific reach: cutting reach to ~ 500 km for interior T/S slots costs no corridor AUC; cutting it for wind/SST slots does.
- 2 Group-specific history: wind slots beyond lag 1 add nothing; interior slots gain from 24 $\rightarrow$ 48 months.
- 3 A cone-shaped stencil (reach $\propto \Delta t + \text{lag}$) matches the 4444 km cylinder at a fraction of the slots.
- 4 The pentad stage-2 shows one-step wind skill the monthly one cannot, because $\Delta t \approx \tau_{\text{atm}}$.

Where the simple picture needs care

v is the fastest relevant signal speed, not the mean flow — waves (Rossby, Kelvin, barotropic) outrun advection. τ is predictability, not causality: the cone is truncated by memory, so wind's 'large area' is bounded by $v\tau$ ($\sim 4,000\text{--}8,000$ km), not by v -horizon. Cones are tilted, not symmetric: upstream along the flow, westward for Rossby waves, along boundaries for coastal waves. Time-averaging to monthly means turns anything with $\tau < \Delta t$ into spatially-correlated forcing — the atmosphere is a stochastic boundary condition at our cadence, not a dynamical state. Eddy diffusion adds only $\sqrt{K\cdot t} \approx 50$ km/month at $K \approx 10^3 \text{ m}^2 \text{ s}^{-1}$ — negligible at $\frac{1}{4}^\circ$.

What the cone predicts — and how to test it on our stencil

The cone model applied channel group by channel group, with the useful reach per monthly step, the useful history, and a design consequence for each.

Wind, pressure and fluxes: the atmosphere moves at 10 m/s and forgets in 5–10 days. Its causal reach in one month is the whole globe, but because its memory is shorter than our step, monthly wind is not predictable from its own history — only through slow ocean modes that force it. So treat it as a wide, shallow forcing: large stencil, no depth. At the five-day step the memory and the step are similar, so wind becomes partly self-predictable — a gain only the pentad model could show.

Sea-surface and mixed-layer temperature: advected at 0.1–0.3 m/s (260–800 km per month), forced by the atmosphere, remembered for 3–12 months (including "re-emergence" — anomalies that sink in autumn and reappear next spring). It needs both the wide wind stencil and months of history: the one group that justifies the full cylinder.

Interior temperature and salinity: 0.02–0.1 m/s, memory of years to decades. Per monthly step the useful reach is 50–300 km, which a 3-by-3 patch at quarter degree already covers; our 4444-km stencil is 15–90 times too wide for one step. Small reach, long history — the concrete answer to the open question "can reach come down?" from the project's dependency-cone note.

Sea-surface height: slow Rossby waves in the interior (0.03 m/s at mid-latitudes), fast Kelvin and coastal waves along boundaries (1–3 m/s). An anisotropic stencil — an arm along coasts, narrow elsewhere.

The transport target (RAPID): the overturning transport is set by the east-west density difference across the basin ("thermal wind"), which means its cone is the whole section — the reason our pooled read-out failed and the unpooled attention head worked.

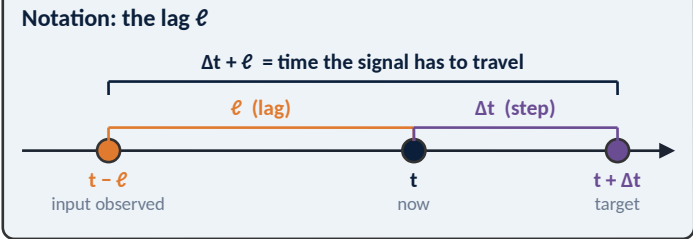
Bottom left: four falsifiable predictions, each an ablation we can run with the existing evaluator. Bottom right: where the simple picture needs care — v is the fastest signal speed, not the mean flow; τ is predictability, not causality; cones are tilted, not symmetric; monthly averaging turns anything faster than the step into forcing.

Worked example — the surface current at one pixel

Target: (u, v) at a $\frac{1}{4}^\circ$ pixel P , one month ahead. Which drivers, how each one moves the current, and what (v, τ) it carries. Live version: the Dependency-Cone Explorer artifact.

Driver	Mechanism → current	How it enters the stencil	v	τ	L_{corr}	Useful lags ($\Delta t = 1$ mo)	Reach: lag 0 → last useful lag
Wind stress	Ekman transport & surface drift — responds within ~1 day, locally	direct forcing; $\tau < \Delta t \Rightarrow$ lag 0 only, not itself predictable	10 m/s	10 d	1,500 km	lag 0 only	10,000 (cap) km
Air temperature / heat flux	net heat flux → mixed-layer T → density gradient → geostrophic current	fast driver integrated by a slow medium: inherits $\tau_{\text{ML}} \approx 6$ mo and, for old lags, ocean advection speed	10 m/s (atm.) 0.15 m/s (ocean)	6 mo	1,500 km	0 – 5 (6 of 24)	1,889 → 3,833 km
Ocean surface T (mixed layer)	horizontal density gradient → thermal wind	via the gradient $\Rightarrow$ neighbours needed ($\geq 3 \times 3$); advected at 0.15 m/s	0.15 m/s	8 mo	—	0 – 7 (8 of 24)	389 → 3,110 km
Interior T, S (below ML)	thermal wind integrated from depth; Rossby adjustment	via the gradient; 0.03 m/s $\Rightarrow$ 78 km per month — the 3×3 patch covers lag 0	0.03 m/s	5 yr	—	0 – 23 (24 of 24)	78 → 1,866 km
SSH	pressure gradient → barotropic geostrophic current	via the gradient; Rossby 0.03 m/s westward + Kelvin arm 2.5 m/s along coasts (anisotropic)	0.03 m/s	1 yr	—	0 – 11 (12 of 24)	78 → 933 km
Own history $(u, v \text{ at } P)$	momentum persistence, mesoscale eddies (westward drift ~0.03 m/s; eddies live 4+ months but pass through a fixed pixel in weeks–months)	direct; eddy memory ≈ 3 months	0.15 m/s	3 mo	—	0 – 2 (3 of 24)	389 → 1,166 km

Two rules do all the work. $\text{reach}(\ell) = \min(\max(v \cdot (\Delta t + \ell), L_{\text{corr}}), 10,000 \text{ km})$ — a signal observed ℓ before now has $\Delta t + \ell$ to travel; a field is never less coherent than its own correlation length. A lag is useful iff $\Delta t + \ell \leq \tau$ (lag 0 is always kept). One exception: a fast driver acting through a slow medium (air temperature → heat flux → mixed layer) adds instead of maxing — $\text{reach} = L_{\text{corr}} + v_{\text{ocean}} \cdot (\Delta t + \ell)$ — because the flux pattern is laid down over L_{corr} and then carried away. Wind is the extreme case: at monthly cadence its memory is shorter than the step, so it enters as one wide sheet of forcing at lag 0 and nothing older carries information about next month's wind. Air temperature is the subtle case: the atmosphere forgets in 10 days, but the mixed layer remembers the heat it delivered for months and carries it along at 0.15 m/s — so the channel inherits the ocean's τ and v .



Worked example — the surface current at one pixel

A concrete walk-through of the cone idea for one target: the surface current (u , v) at a quarter-degree pixel, one month ahead. Each row is a driver — something that pushes the current around — with its mechanism and its two numbers, speed and memory, plus a correlation length.

Wind stress acts through Ekman transport (the wind drags the surface layer, which turns at right angles because of Earth's rotation), within about a day. Because its 10-day memory is shorter than the monthly step, only the newest input (lag 0) is useful, and only as forcing that is not itself predictable.

Air temperature acts through heat flux, which changes mixed-layer temperature, which changes density gradients, which drive geostrophic currents. This is the subtle row: the atmosphere forgets in 10 days, but the mixed layer remembers the heat it received for months and carries it along at ocean speed — so the channel inherits the ocean's memory (6 months) and speed (0.15 m/s). Its reach is the 1,500-km atmospheric correlation length plus what the ocean advects.

Ocean surface temperature drives currents through horizontal density gradients ("thermal wind") — which is why neighbours are mandatory: a gradient does not exist inside one pixel.

Interior temperature and salinity: same mechanism, integrated from depth; at 0.03 m/s (78 km per month) the 3-by-3 patch already covers lag 0.

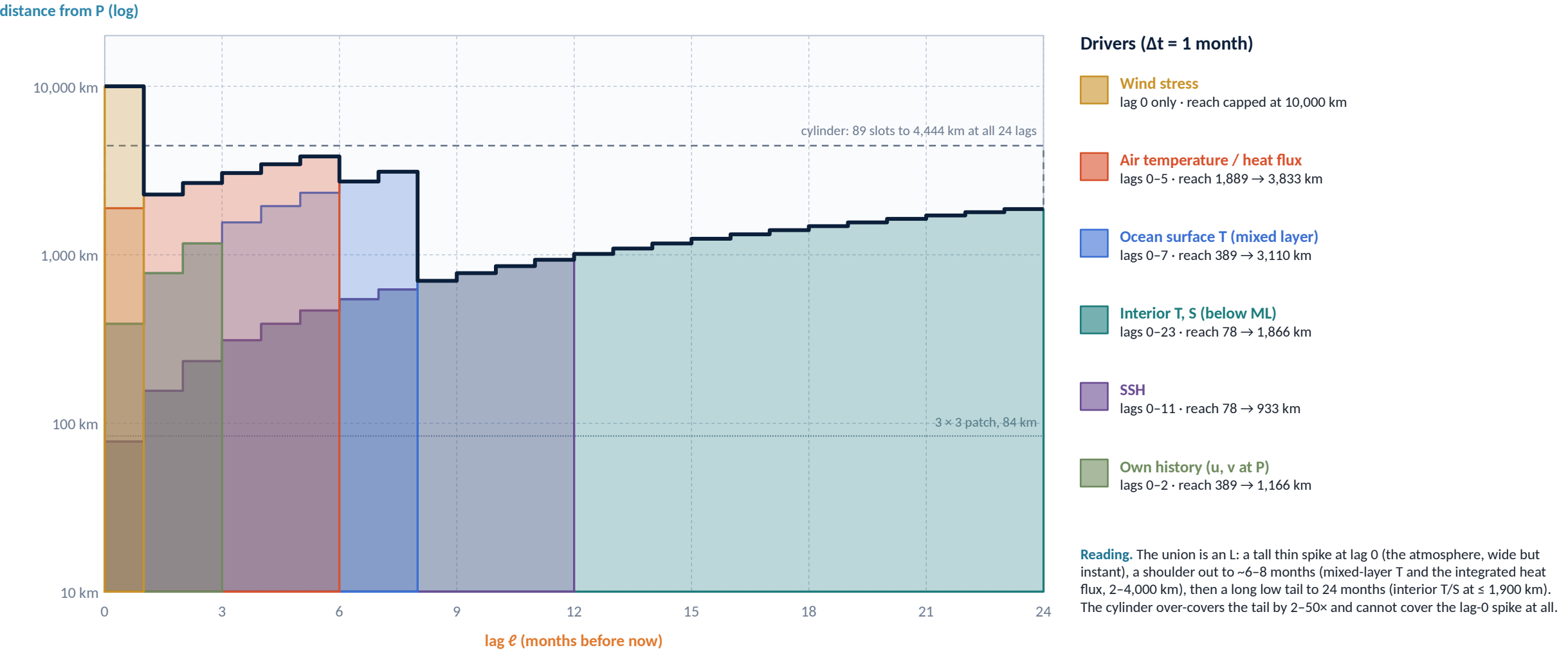
Sea-surface height sets the pressure gradient; slow Rossby waves in the interior plus a fast Kelvin arm along coasts.

Own history: momentum persistence and eddies. Individual eddies live four months and longer and drift westward at a few centimetres per second, but at a fixed pixel the current decorrelates in weeks to a few months because they pass through — that decorrelation time, about three months, is the memory that matters here.

The "two rules" box states the arithmetic used throughout: reach at lag ℓ is the larger of $v \cdot (\Delta t + \ell)$ and the correlation length, capped at 10,000 km; a lag is useful if $\Delta t + \ell \leq \tau$. There is one deliberate exception, stated in the box: for a fast driver acting through a slow medium (air temperature $\rightarrow$ heat flux $\rightarrow$ mixed layer) the two lengths add rather than compete — the flux pattern is laid down over the atmospheric correlation length and then carried away by the ocean — which is why the air-temperature row reads 1,889 to 3,833 km. The notation panel defines ℓ with a timeline. All parameter values are order-of-magnitude illustrations and can be changed live in the Dependency-Cone Explorer page.

Step 1 — each driver becomes a wedge in (lag, distance)

x: how long before now the input was observed · y: how far from P it may sit and still matter (log). Bold outline = the union the stencil should follow.



Step 1 — each driver becomes a wedge in (lag, distance)

Each driver from the previous slide becomes a wedge on this chart. Horizontal axis: lag ℓ , how many months before now the input was observed (0 to 24). Vertical axis (logarithmic): how far from the pixel the input may sit and still matter, 10 km to 10,000 km.

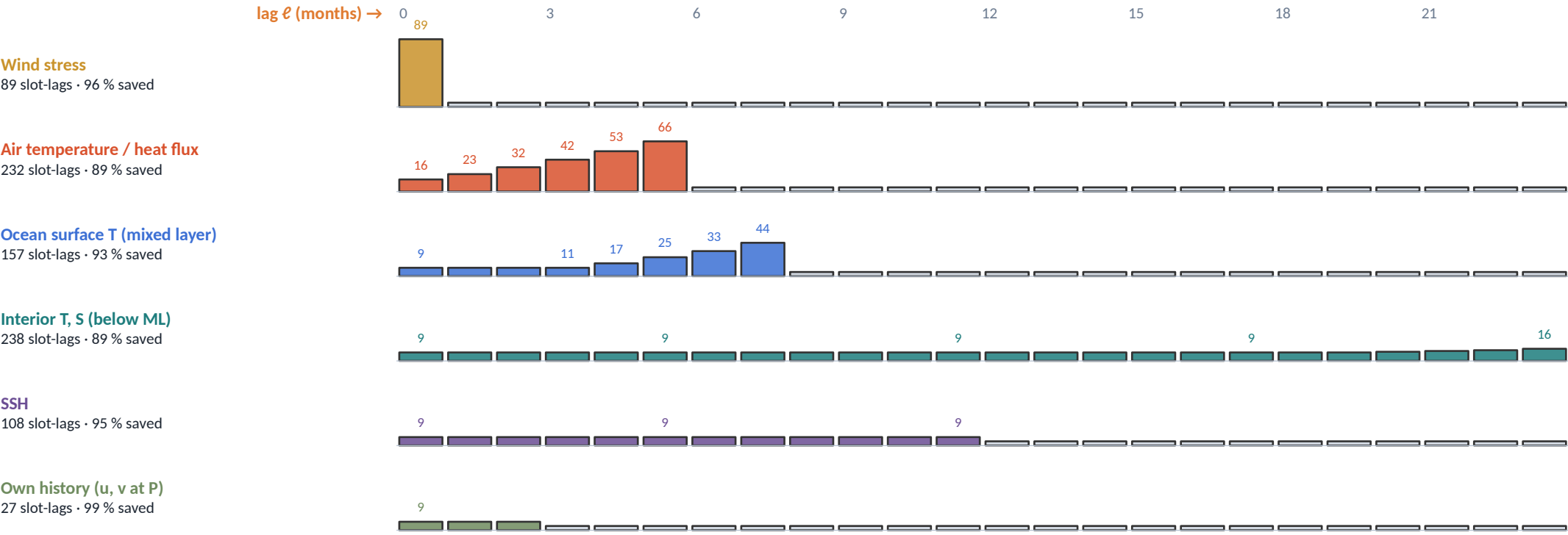
A wedge rises to the right because older inputs have had more time to travel — reach grows as $v \cdot (\Delta t + \ell)$ — and stops abruptly where the driver's memory τ runs out. Wind is a tall spike at lag 0 only. Air temperature is a wide band for six lags. Surface temperature is a true wedge for eight lags. Interior temperature and salinity and sea-surface height are low and long. Own history is short.

The bold black line is the union of all wedges: the region a stencil for this target should follow. Its shape is an L: a spike at lag 0 (the atmosphere, wide but instant), a shoulder out to 6–8 months (mixed-layer temperature and integrated heat flux, 2,000–4,000 km), then a long low tail to 24 months (the interior, under 1,900 km).

Two reference lines: the dashed rectangle is our current "cylinder" — 89 stencil slots reaching 4,444 km at every one of 24 lags; the dotted line at 84 km is the 3-by-3 patch. The cylinder over-covers the tail by a factor of 2 to 50 and cannot cover the lag-0 spike at all.

Step 2 — the union becomes a per-channel, lag-dependent stencil

Slots needed at each lag = $89 \cdot (\text{reach} / 4,444 \text{ km})^2$, floored at the 9-slot 3×3 patch and capped at 89. Same 89-slot sunflower, different footprint per channel and lag.



851

cone-shaped slot-lags for the six channels

vs

12,816

for the cylinder ($89 \times 24 \times 6$)

→

93.4 % fewer

Bar height = slots needed at that lag (max 89). Grey stubs = lags outside the driver's memory: dropped. Numbers label the first lag, every sixth, and the first few changes.

What the shapes say. Wind is one full-width sheet at lag 0 and nothing else. Air temperature is a wide sheet for six lags (the mixed layer's memory). Surface T is a true cone: 9 slots at lag 0 growing to 44 at lag 7. Interior T/S is a thin 24-lag column at the 9-slot floor for a year and a half before it widens. SSH is a thin 12-lag column (plus the coastal arm the isotropic count ignores). Own history is three short lags. The cylinder spends 2,136 on each of them; the budget frees ~93 % of the slots for the ones that matter — or lets us afford 48-month history for the interior channels at no extra cost.

Step 2 — the union becomes a per-channel, lag-dependent stencil

This slide turns the wedges into a stencil budget. Our production stencil, the "sunflower-89", spreads 89 slots over a disc of radius 4,444 km and is applied at all 24 lags: $89 \times 24 = 2,136$ "slot-lags" per channel — the cylinder.

The rule: slots needed for a given reach r scale with area, $89 \cdot (r/4,444 \text{ km})^2$, floored at 9 (the 3-by-3 patch) and capped at 89. Each row is a driver; each bar is a lag; bar height is the slots needed at that lag; grey stubs are lags outside the driver's memory, which are dropped.

Wind: one full-width sheet at lag 0, nothing else — 89 slot-lags. Air temperature: a wide sheet for six lags, 232. Surface temperature: a true cone from 9 to 44 slots over eight lags, 157. Interior temperature and salinity: a thin 24-lag column at the 9-slot floor for a year and a half before it widens, 238. Sea-surface height: a thin 12-lag column, 108 (its coastal arm is not counted by this isotropic formula). Own history: three short lags, 27.

Total: 851 cone-shaped slot-lags for the six channels against 12,816 for the cylinder — 93 % fewer. Equivalently, the saving could buy 48 months of history for the interior channels at no extra cost. These numbers follow directly from the illustrative speeds and memories; the Explorer page recomputes them if you change any parameter.

Step 3 — the physics behind each wedge

The stencil fixes the support; attention learns the weights. Four ablations on this target decide whether the support is right.

Ekman — wind → drift

Surface stress τ drives a transport $\approx \tau / (\rho f)$ at 90° to the wind within ~ 1 inertial period (≈ 1 day at 30°). Local, fast, shallow (tens of m). At monthly cadence this is same-step forcing: the wind of month $t+1$ is unknown, the wind of month t is already in the ocean state.

Thermal wind — T, S gradients → shear

$\partial u / \partial z = (g / (f \rho_0)) \partial \rho / \partial y$: horizontal density gradients set the vertical shear of the geostrophic current, integrated from depth. A gradient does not exist inside one pixel — this is why the 3×3 patch is the floor for T/S channels and why heat flux matters through its spatial pattern, not its local value.

Pressure gradient — SSH → surface geostrophy

$u_g = -(g/f) \partial \eta / \partial y$. SSH anomalies arrive as slow westward Rossby waves in the interior (0.03 m/s) and as fast Kelvin/coastal waves along boundaries (2.5 m/s): a thin column with one long arm — the one wedge the isotropic count under-states.

Advection & eddies — own history

Momentum persists and mesoscale eddies drift westward at ~ 0.03 m/s. Individual eddies live 4 months and more (Chelton et al. 2011), but at a fixed pixel the velocity decorrelates in weeks to a few months because they pass through — hence $\tau \approx 3$ months for own history. Upstream cells at 0.15 m/s $\times (\Delta t + \ell)$ carry what will arrive. Short cone, tilted upstream.

What the stencil fixes vs what attention learns

The cone is the support — which (cell, lag, channel) tokens may enter at all. It encodes only two physical facts per driver: how fast its signal travels and how long it is predictable. Everything else — which upstream cell matters this month, how strongly the SSH gradient projects onto u , whether the heat-flux pattern is the one that tilts the density field — is left to the attention weights inside the support. A support that is too small cannot be learned around; one that is too large costs slots and lets noise in. The four ablations to the right test whether the support is right, run with the existing 12-month rolled corridor-AUC evaluator and the frozen gate head.

The four ablations, specialised to this target

- 1 Interior T/S slots at 500 km reach instead of 4,444 km: predicted no loss (their wedge never exceeds 1,900 km).
- 2 Wind and heat-flux slots restricted to lags 0–1: predicted no loss ($\tau_{\text{atm}} < \Delta t$); the same cut on surface T: predicted loss (its wedge runs to lag 7).
- 3 Cone-shaped stencil (851 slot-lags) vs cylinder (12,816): predicted equal corridor AUC at 7 % of the slots — and headroom for 48-month interior history.
- 4 Pentad codec ($\Delta t = 5$ d $\approx \tau_{\text{atm}}$): predicted one-step wind skill that the monthly model cannot show, because wind's own wedge now has lags > 0 .

Two codec seeds minimum; head numbers are never quoted from one. Explore the parameters live in the Dependency-Cone Explorer artifact (same formulas, editable v , τ , L_{corr} , Δt , anisotropy).

Step 3 — the physics behind each wedge

The physics behind each wedge, in four cards, and then what is fixed versus what is learned.

Ekman: wind stress τ drives a surface transport of about $\tau/(\rho f)$ at right angles to the wind — ρ is water density, f the Coriolis parameter set by latitude — within about one inertial period, roughly a day at 30° . Local, fast, shallow. At monthly cadence this is same-step forcing.

Thermal wind: the vertical shear of the geostrophic current is proportional to the horizontal density gradient, $\partial u/\partial z = (g/(f \rho_0)) \partial \rho/\partial y$. A gradient does not exist inside one pixel, which is why the 3-by-3 patch is the floor for temperature and salinity channels, and why heat flux matters through its spatial pattern rather than its local value.

Pressure gradient: the surface geostrophic current is $u = -(g/f) \partial \eta/\partial y$, where η is sea-surface height. SSH anomalies arrive slowly as westward Rossby waves in the interior and fast as Kelvin or coastal waves along boundaries.

Advection and eddies: momentum persists and mesoscale eddies drift westward at a few centimetres per second. The eddies themselves live four months and more, but at a fixed pixel the current decorrelates in weeks to a few months as they pass through — that is the memory used for "own history". Upstream cells carry what will arrive.

What the stencil fixes versus what attention learns: the cone is the support — which (cell, lag, channel) tokens may enter at all. It encodes only speed and memory per driver. Everything else — which upstream cell matters this month, how strongly an SSH gradient projects onto the current — is learned by the attention weights inside the support. A support too small cannot be learned around; one too large costs slots and admits noise.

Right box: the four ablations specialised to this target, with the predicted outcomes. They are hypotheses; nothing here has been run. The Dependency-Cone Explorer page uses the same formulas with editable parameters.

Related work I — the dependency cone vs. the literature

Who has used 'speed × time' to size what a model sees, and who has not (survey Aug 2026)

Work	What it does	Shares with the cone idea	Does not
Courant, Friedrichs & Lewy 1928 — domain of dependence	Numerical PDE stability: the numerical stencil must contain the physical domain of dependence $\Delta x \leq v \Delta t$.	The origin of the argument; our ' $\Delta x \leq v \Delta t$ ' is exactly this	No notion of memory τ ; about stability, not information
OceanNet — Chattopadhyay et al. 2024, Sci. Rep. (arXiv 2310.00813)	Regional FNO for Gulf Stream SSH, 4–5-day mean lead, 80 M params; justifies dropping lateral boundary conditions: 'Rossby waves at 40°N $\approx$ 1 cm/s ... eddies travel only 30–120 km from the open boundary' over the horizon.	Explicit wave-speed × time argument in an ocean ML model	Used to shrink a boundary, not to shape a per-channel stencil; single process
PARADIS 'Learning to Advect' — Pereira et al. 2026 (arXiv 2601.21151)	Neural semi-Lagrangian layer: gathers features at the departure point $x_d = x - \Delta t \cdot u(x)$; stability becomes a Lipschitz condition on u instead of CFL; a 20-parameter NSL layer moves a tracer a 3-layer U-Net's receptive field cannot.	Input support set by velocity × Δt , learned per point — the anisotropic, tilted cone made operational	One transport field for all variables; no τ truncation; atmosphere only
ClimODE — Verma et al., ICLR 2024	Neural ODE with the continuity/advection equation; learns a velocity field, but reads inputs with fixed 3×3 convs plus a global-attention branch.	Advection as inductive bias	Local/global mix fixed, not sized by $v \Delta t$ per variable
Limited-area ML weather: Neural-LAM (Oskarsson 2023/24), stretched grid (Nipen 2024), Diffusion-LAM 2025	Boundary forcing zone of a fixed 10 grid nodes (~100 km) at every step, or a smooth global→regional mesh.	A boundary halo is a cone cut by the domain edge — our 'window edge' problem	Halo width is static; not derived from wind speed × lead time
Global ML weather: Keisler 2022 (9 GNN hops / 6 h), GraphCast (multi-mesh, 16 layers), Pangu (1/3/6/24 h chain), FuXi (0–5/5–10/10–15 d cascade)	Per-step reach is set by hop count or window size; longer horizons handled by bigger steps or horizon-specific models.	Implicit speed bound per step; growing the cone with lead time by changing Δt	Same reach for every channel; no receptive-field-vs-lead-time ablation published
Per-variable tokenisation: ClimaX 2023, Aurora 2024/25, Prithvi WxC 2024, Stormer	Each variable gets its own embedding / patch tokens, aggregated by attention.	Variable-specific processing as a design pattern	Variable-specific embedding, not variable-specific spatial or temporal extent
Ocean emulators: Samudra 2024/25 (1°, 19 levels, 5-day step, ConvNeXt, atmosphere prescribed), ACE2-SOM (mirror: ML atmosphere + slab ocean), Xihe, GLONET	One medium is treated as prescribed forcing at a cadence set by the other's dynamics.	'Atmosphere as forcing' at slow cadence — our monthly- Δt inversion in practice	Not stated as a τ -vs- Δt rule; no channel-specific stencils
Earth 2 dependency-cone note (16 Aug) + this deck	Cone recursion proved bit-equal to a full roll with NaN poisoning; reach 222 vs 4444 km per month measured as 29.5 % vs 100 % of the world ocean after 12 months; now: per-process (v , τ) cones, union rule, channel-group stencils.	—	Open: the τ -truncation + union → per-channel stencil principle was not found stated anywhere in our search; it is a hypothesis to test (four ablations), not a result

Reading. The 'speed × time' half of the idea has two clear precedents — OceanNet's Rossby-speed boundary argument and PARADIS's semi-Lagrangian gather — and the 'one medium as forcing' half is standard practice in ocean emulators. What we did not find is the combination: truncating each process's cone by its own memory, taking the union per channel, and letting that dictate a channel-specific stencil shape. Absence in a one-day search is not absence in the literature; treat it as 'not yet found', and cite OceanNet and PARADIS as the nearest neighbours.

Related work I — the dependency cone vs. the literature

Where does the dependency-cone idea stand against the literature? Row by row.

The origin is the Courant–Friedrichs–Lewy condition from 1928: a numerical scheme's stencil must contain the physical domain of dependence, $|\Delta x| \leq v \cdot \Delta t$. Our first rule is exactly this; the CFL condition has no notion of memory.

OceanNet (2024), a regional neural forecaster for the Gulf Stream, uses precisely a wave-speed-times-time argument to justify dropping lateral boundary conditions: Rossby waves at 40°N move about 1 cm/s, so eddies travel only 30–120 km from the open boundary over the forecast horizon. Same arithmetic, used to shrink a boundary rather than to shape a per-channel stencil.

PARADIS (2026) is the architectural version: a "neural semi-Lagrangian" layer gathers features at the departure point $x - \Delta t \cdot u$ — where the flow came from — turning the CFL limit into a smoothness condition on the velocity. A 20-parameter layer moves a tracer that a U-Net's fixed receptive field cannot. That is the tilted, anisotropic cone made operational, though with one transport field for all variables and no memory truncation.

ClimODE uses advection as an inductive bias but with fixed local and global branches. Limited-area weather models use fixed boundary halos, not sized by wind speed times lead time. Global models such as GraphCast, Pangu and FuXi set reach per step by hop count or window size and grow it with lead time by changing the step — the same reach for every channel. Per-variable tokenisation (ClimaX, Aurora, Prithvi WxC) gives each variable its own embedding, not its own spatial or temporal extent. Ocean emulators such as Samudra prescribe the atmosphere as forcing — our monthly-step inversion in practice.

What we did not find is the combination: truncating each process's cone by its own memory, taking the union per channel, and letting that dictate a channel-specific stencil. A one-day search is not a literature review; treat it as "not yet found", and cite OceanNet and PARADIS as the nearest neighbours.

Related work II — the four-sphere plan vs. the literature

Coupled emulators, observation-native models, 'Earth-system' foundation models, biosphere ML, coupled data assimilation (survey Aug 2026)

Family • nearest works	What they do	Shares with our plan	Does not
Coupled emulators — Samudra → SamudrACE (Dheeshjith 2025; Duncan 2026), ACE2-SOM (2025), ACE2-NEMO (2026)	Samudra: 3-D global ocean emulator (T, S, currents at depth, 1°, 5-day step); SamudrACE couples it to Ai2's ML atmosphere ACE2. ACE2-SOM/NEMO couple an ML atmosphere to a slab / dynamical ocean.	SamudrACE is the closest structural analogue: an ocean-interior model first, atmosphere joined later	Emulators of a simulator, not embeddings of observations; no ocean colour, no land biosphere
AIFS surface ocean — Hahner, Zampieri et al., ECMWF 2026 (arXiv 2604.25559)	One encoder–processor–decoder forecasts atmosphere + SST, SSS, SSH, surface currents, waves, sea ice 'in a component-agnostic way'; slowly evolving ocean/ice fields get larger loss-scaling factors.	Joint multi-sphere state space; explicit handling of the timescale mismatch	Surface ocean only; forecast model, no reusable embedding; no biosphere
Aurora — Bodnar et al., Nature 2025 ('A foundation model for the Earth system'); Aurora 1.5	One pretrained 3-D Swin backbone, fine-tuned to air quality, ocean waves, cyclones.	One representation, many sphere-specific heads	Atmosphere-centric; no interior ocean, no two-way coupling, no biosphere. Note the title collision
Observation-native — Aardvark Weather (Allen et al., Nature 2025), GraphDOP (ECMWF 2024/25), 4DVarNet (Fablet et al. 2023/24)	Aardvark/GraphDOP learn forecasts or a latent state directly from raw observations, skipping reanalysis; 4DVarNet learns a variational solver that fuses sparse altimetry + SST into gap-free surface fields.	The recipe for taking Argo profiles, swaths and tracks natively — what 'combine all relevant data' needs	Atmosphere (Aardvark, GraphDOP) or 2-D surface ocean (4DVarNet); no Argo-depth encoder found
'Earth-system' FMs — ESFM (Ozdemir et al. 2026, arXiv 2605.00850), Copernicus-FM (2025), Prithvi family (EO, WxC, Surya), AlphaEarth (2025)	ESFM: Aurora-style backbone with per-variable tokens over ERA5, CMIP6, sparse MODIS and station data with missingness handling. Copernicus-FM unifies Sentinel sensors incl. S-5P. Prithvi: separate sphere-specific FMs. AlphaEarth: embedding field with ERA5-Land as an input.	Heterogeneous sources in one embedding; missingness as a first-class token (ESFM) — same primitive as our masked / never-measured tokens	All atmosphere- or land-centric; none has the ocean interior or the ocean biosphere as a target
Ocean biosphere / carbon ML — SOM-FFN (Landschützer), CSIR-ML6 (Gregor 2019), pCO ₂ -Residual (Bennington 2022), CANYON-B (Sauzède 2017), Granite-Geospatial-Ocean (2025)	Data-driven surface-pCO ₂ and carbon-sink maps from SST, SSS, chl, MLD; NN inference of nutrients/carbonate from T, S, O ₂ for BGC-Argo; one ocean-colour FM.	Exactly the physics → biology couplings step 2 of our sequence would learn end-to-end	Task models with hand-chosen predictors; no shared representation with the physical state
Coupled data assimilation — Penny & Hamill 2017 (BAMS); Penny et al. 2019 (JAMES); ECMWF CERA	The canonical statement that atmosphere (days) and ocean (months–years) memories differ by orders of magnitude; strongly vs weakly coupled DA; production systems still weakly coupled.	The multi-timescale problem is known and hard; our cone-per-sphere is a representation-side answer to it	Assimilation, not learned representation
Platforms & twins — NVIDIA Earth-2, Destination Earth (ECMWF/CMCC/ESA), EU Digital Twin of the Ocean (EDITO)	Generative km-scale weather tooling (NVIDIA); physics-based coupled km-scale twins (DestinE); ocean data/model platform (EDITO).	Same four-sphere integration goal at production scale	Not learned embeddings; NVIDIA 'Earth-2' is a name collision we must disambiguate

Reading. The nearest lineage on the physics side is Samudra → SamudrACE (ocean interior first, atmosphere second — but emulating a simulator); on the representation side it is ESFM and Copernicus-FM (heterogeneous sources, missingness tokens — but no ocean). Nobody starts from the ocean interior as an embedding, and nobody joins the ocean biosphere to a physical-state representation. Two names to disambiguate in any write-up: NVIDIA's 'Earth-2' platform and Aurora's 'A foundation model for the Earth system'.

Related work II — the four-sphere plan vs. the literature

The four-sphere ambition against the literature, by family.

Coupled emulators: Samudra is a 3-D global ocean emulator (temperature, salinity, currents at depth) trained on reanalysis; SamudrACE (2026) couples it to Ai2's machine-learned atmosphere ACE2. This is the closest structural analogue to our order — ocean interior first, atmosphere joined later — but it emulates a simulator rather than embedding observations, and has no biosphere. ACE2-SOM and ACE2-NEMO couple an ML atmosphere to a slab or dynamical ocean.

ECMWF's 2026 AIFS surface-ocean paper forecasts atmosphere plus sea-surface temperature, salinity, height, currents, waves and sea ice in one "component-agnostic" model, and explicitly gives the slowly evolving ocean and ice fields larger loss weights — the first ML system we found that names and handles the cross-sphere timescale mismatch. Surface only, and a forecast model rather than a reusable embedding.

Aurora (Nature 2025, titled "A foundation model for the Earth system") is one pretrained backbone with sphere-specific fine-tunes; atmosphere-centric, no interior ocean.

Observation-native learning: Aardvark Weather and GraphDOP learn directly from raw observations, skipping reanalysis; 4DVarNet learns to fuse sparse altimetry and temperature into gap-free surface fields. This is the recipe for taking Argo profiles, swaths and ship tracks natively — what "combine all relevant data" needs — but none of them has an Argo-depth encoder.

"Earth-system" foundation models: ESFM (2026) and Copernicus-FM unify heterogeneous sources with missingness tokens — the same primitive as our "masked" versus "never measured" tokens — but neither touches the ocean. The Prithvi family is a set of separate sphere-specific models; AlphaEarth uses ERA5-Land only as an input.

Ocean-biosphere and carbon ML (SOM-FFN, CSIR-ML6, CANYON-B, Granite) are task models with hand-chosen predictors. Coupled data assimilation (Penny & Hamill 2017) is where the timescale problem was first named. Platforms and twins (NVIDIA Earth-2, Destination Earth, EDITO) share the goal but are not learned embeddings — and NVIDIA's "Earth-2" is a name collision we must disambiguate, as is Aurora's title.

Sources

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TESSERA arxiv.org/abs/2506.20380 · github.com/ucam-eo/tessera · github.com/ucam-eo/geotessera · geotessera.org · CVPR 2026 open-access page

OlmoEarth arxiv.org/abs/2511.13655 · allenai.org/blog/olmoearth-models · github.com/allenai/olmoearth_pretrain · huggingface.co/blog/allenai/olmoearth-v1-1 · allenai.org/papers/olmoearth-v1-2

TerraMind arxiv.org/abs/2504.11171 · arxiv.org/abs/2504.11172 (TerraMesh) · github.com/IBM/terramind · huggingface.co/ibm-esa-geospatial · philab.esa.int (tiny/small release)

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Granite-Geospatial-Ocean arxiv.org/abs/2509.21273 · github.com/ibm-granite/geospatial (granite-geospatial-ocean: config + prithvi_vit_S3.py) · huggingface.co/ibm-granite/granite-geospatial-ocean · research.ibm.com/blog/oceans-AI-model · hartree.stfc.ac.uk case study (Jul 2026) · pml.ac.uk news

Ocean-including models OceanSAR-1: huggingface.co/galeio-research/OceanSAR-1 · arxiv.org/abs/2504.06962 · OceanSAR-2: arxiv.org/abs/2601.07392 · Copernicus-FM: arxiv.org/abs/2503.11849 · huggingface.co/datasets/wangyi111/Copernicus-Pretrain · github.com/zhu-xlab/Copernicus-FM · Aurora: microsoft.github.io/aurora · wekeo.copernicus.eu (ocean FM workspace)

Benchmarks & head-to-head tables OlmoEarth Tables 2, 3, 7 (arxiv.org/abs/2511.13655) · TerraMind Table 6 / PANGAEA (arxiv.org/abs/2504.11171; arxiv.org/abs/2412.04204) · Prithvi-EO 2.0 Tables A1–A4 / GEO-Bench (arxiv.org/abs/2412.02732; arxiv.org/abs/2306.03831) · TESSERA Table 1 (arxiv.org/abs/2506.20380) · GEO-Bench-2 ranks (arxiv.org/abs/2511.15658) · AlphaEarth & TESSERA for LCZ mapping (arxiv.org/abs/2606.20034) · Granite-Ocean Table 1 (arxiv.org/abs/2509.21273) · OceanSAR-1 Tables 1–3 (arxiv.org/abs/2504.06962)

Related work OceanNet arxiv.org/abs/2310.00813 · PARADIS arxiv.org/abs/2601.21151 · ClimODE arxiv.org/abs/2404.10024 · Neural-LAM arxiv.org/abs/2309.17370 · stretched grid arxiv.org/abs/2409.02891 · Keisler arxiv.org/abs/2202.07575 · GraphCast arxiv.org/abs/2212.12794 · Pangu arxiv.org/abs/2211.02556 · FuXi arxiv.org/abs/2306.12873 · Samudra arxiv.org/abs/2412.03795 · SamudrACE doi [10.1029/2025GL119340](https://doi.org/10.1029/2025GL119340) · ACE2-SOM arxiv.org/abs/2412.04418 · ACE2-NEMO arxiv.org/abs/2603.28704 · AIFS surface ocean arxiv.org/abs/2604.25559 · Aurora doi [10.1038/s41586-025-09005-y](https://doi.org/10.1038/s41586-025-09005-y) · Aardvark doi [10.1038/s41586-025-08897-0](https://doi.org/10.1038/s41586-025-08897-0) · GraphDOP arxiv.org/abs/2412.15687 · 4DVarNet doi [10.1029/2023MS003609](https://doi.org/10.1029/2023MS003609) · ESFM arxiv.org/abs/2605.00850 · Penny & Hamill 2017 BAMS · CSIR-ML6 doi [10.5194/gmd-12-5113-2019](https://doi.org/10.5194/gmd-12-5113-2019) · CANYON-B doi [10.3389/fmars.2017.00128](https://doi.org/10.3389/fmars.2017.00128)

Also checked Prithvi WxC (arxiv.org/abs/2409.13598) — atmosphere-only, no ocean downstream tasks found; NASA marine-debris work — not Prithvi-based. Earth 2 numbers quoted from paper/paper.tex (attribution matrix, probe ladder).

Sources

Every paper, model card and repository the deck draws on, grouped by model and topic, as short URLs. A few notes on how they were used.

For the six model slides, numbers were taken from the papers and, where possible, verified against the published configuration files: Prithvi-EO 2.0's token sizes come from its Hugging Face config.json (16-pixel patches for the 300 M model, 14 for the 600 M), and Granite-Geospatial-Ocean's architecture from the config and model file in the ibm-granite GitHub repository (42-pixel tiles, 17 channels, embedding 512, 12 layers, patch size 2 by 2, one frame).

The head-to-head tables were transcribed from the papers' own tables: OlmoEarth Tables 2, 3 and 7; TerraMind Table 6; TESSERA Table 1; Granite Table 1; OceanSAR-1 Tables 1–3; GEO-Bench-2 for independent ranks; and an independent Local Climate Zone study for AlphaEarth versus TESSERA.

For the ocean-including models and related work, sources were checked in August 2026 and include several 2026 papers (AIFS surface ocean, ESFM, SamudrACE, PARADIS, ACE2-NEMO). Earth 2 numbers — the attribution matrix and probe ladder — are quoted from the project's own paper draft.

Items listed as "also checked" are things we looked at and excluded, with the reason, so that nobody re-chases them.